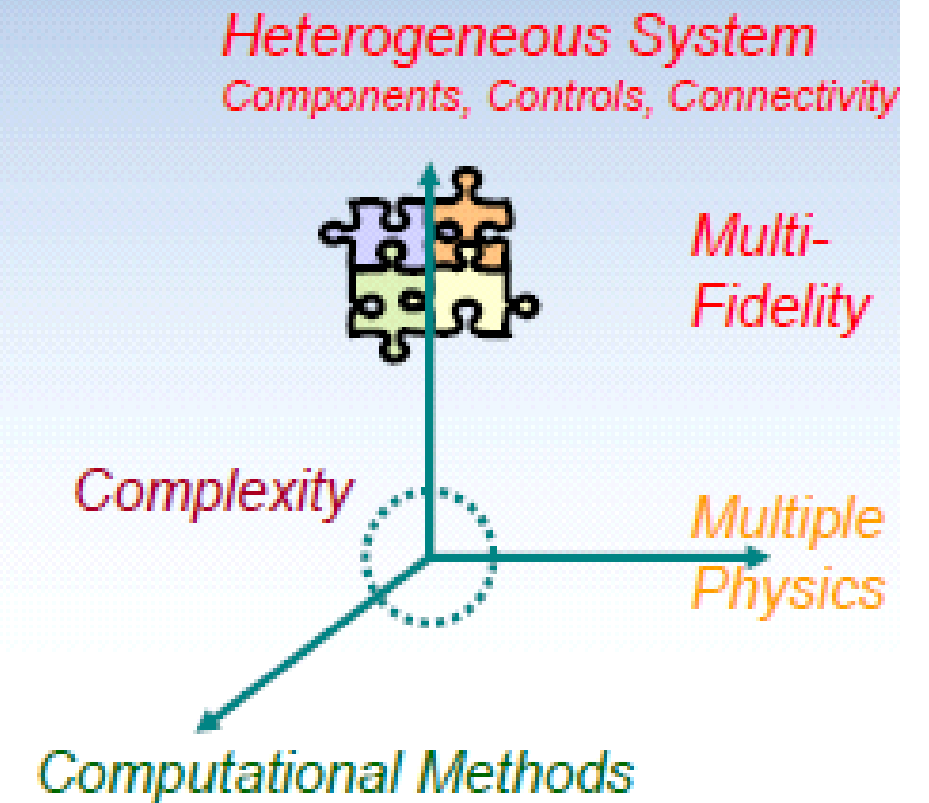
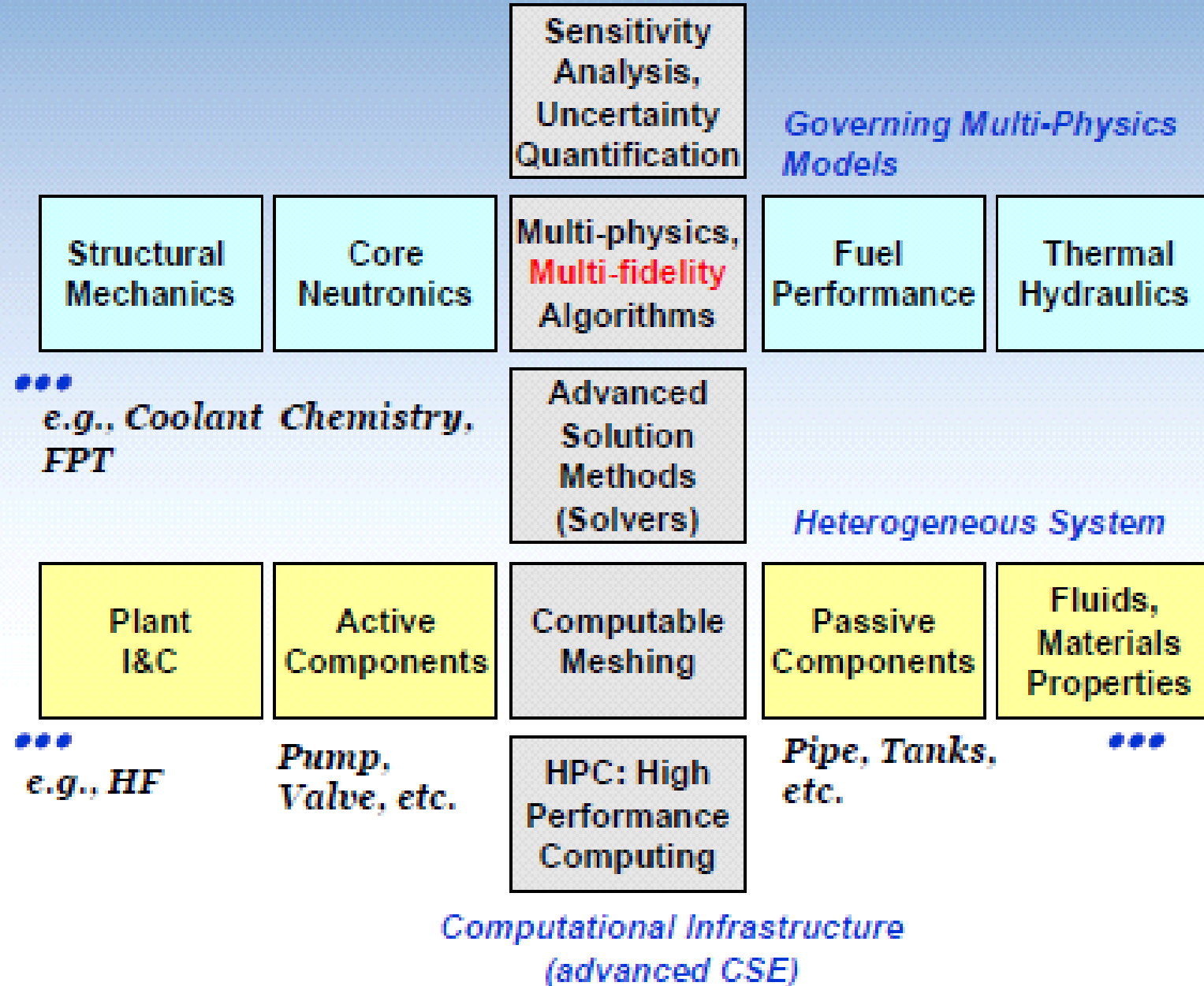


NE 511: Multi-Physics of Nuclear Reactors

UNIT 5: Novel Multi-Physics Modeling

Lecture 20: SPATIAL & TEMPORAL COUPLING - MULTI-PHYSICS PLATFORMS

MULTI-DISCIPLINARY APPROACH TO MULTI-PHYSICS PROBLEM



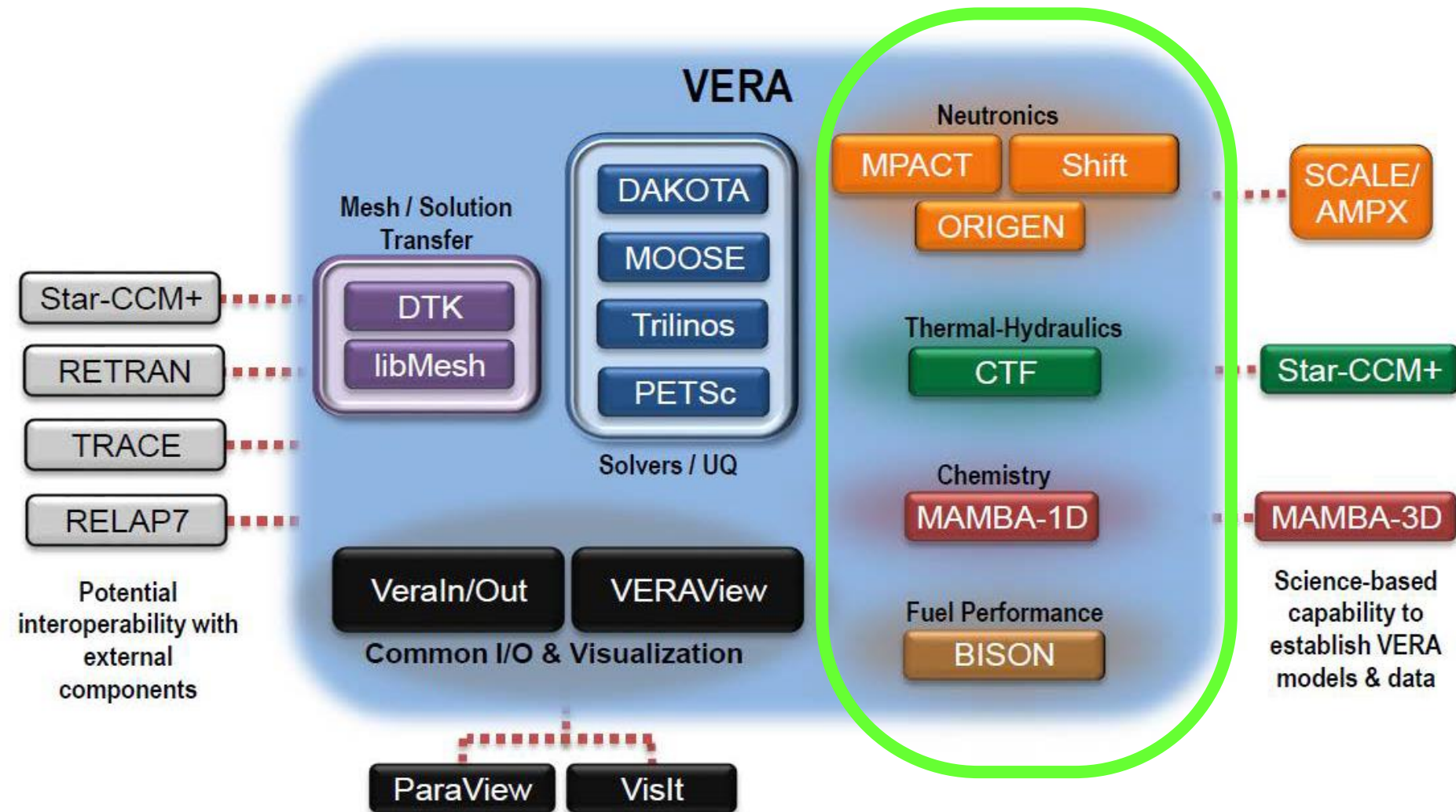
EXAMPLES OF EXISTING MULTI-PHYSICS PLATFORMS

- Multi-Physics Object Oriented Simulation Environment (MOOSE) – a finite element framework for the development of tightly coupled multi-physics solvers;
- SALOME – an open-source integration platform for numerical simulations;
- Virtual Environment for Reactor Applications (VERA), which includes:
 - *LIME (Lightweight Integrating Multi-physics Environment for coupling codes)*
 - *DTK (Data Transfer Kit),*
 - *Trilinos*
 - *Tribal Build, Integrate, and Test System (TriBITS) solvers*
- OpenFoam

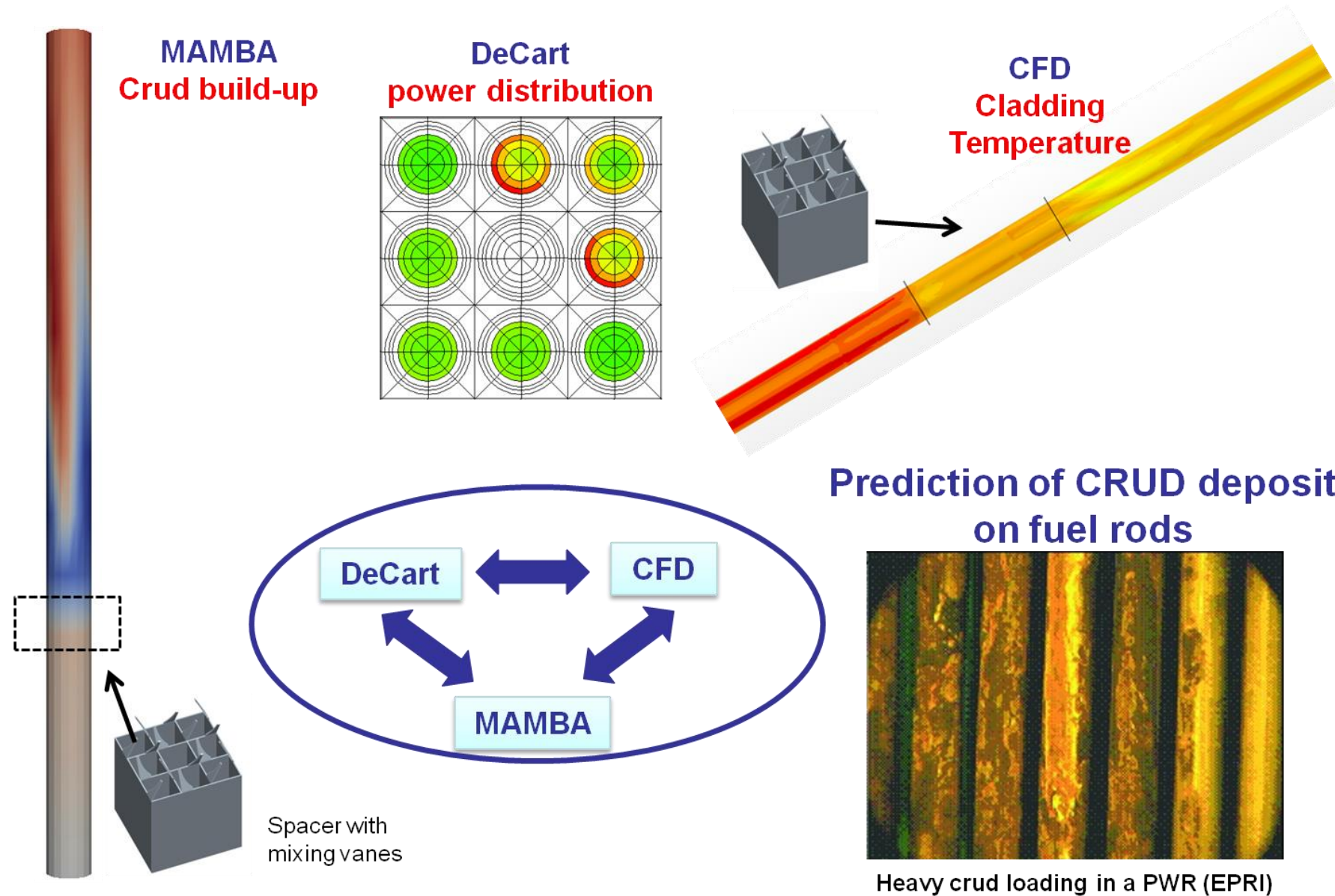
VERA: Virtual Environment for Reactor Applications

CASL's evolving virtual reactor for in-vessel LWR phenomena

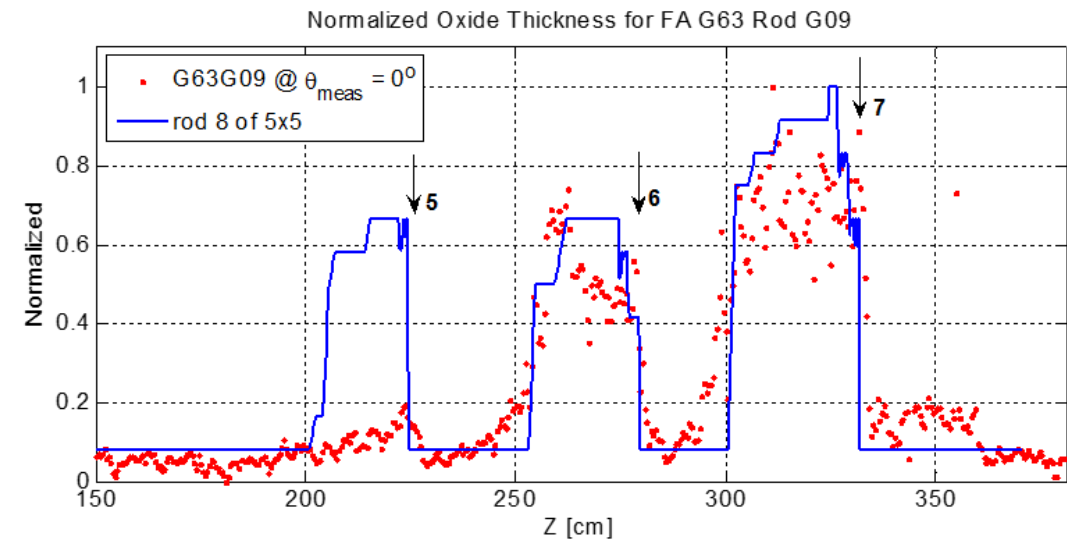
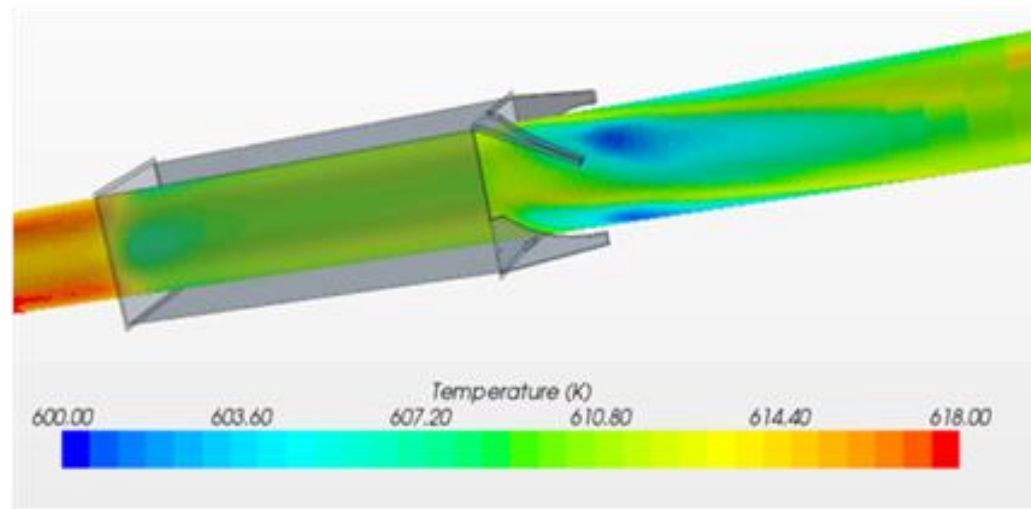
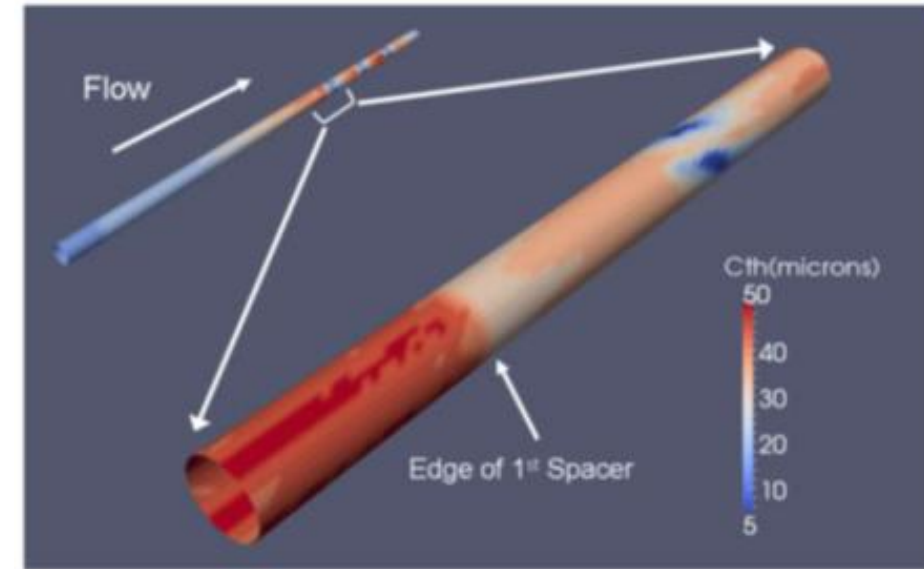
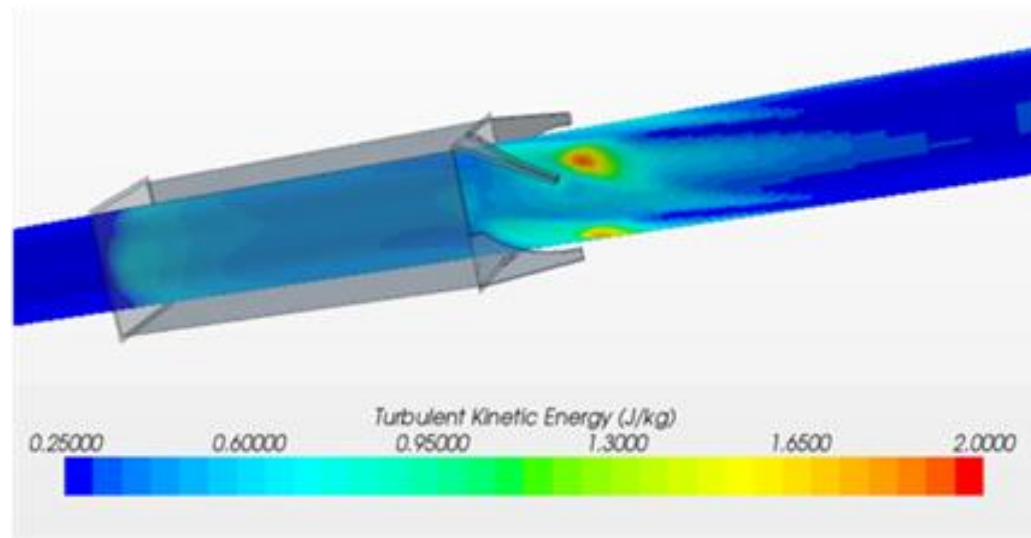
- The DoE established the Consortium of Advanced Simulation of Light Water Reactors (CASL) in 2010.
- CASL developed VERA for high fidelity pin-by-pin reactor modeling and simulation under steady state and depletion conditions.
- Many studies and V&V activities have been conducted.



Multi-physics coupling and data exchange for high fidelity CRUD deposition simulations

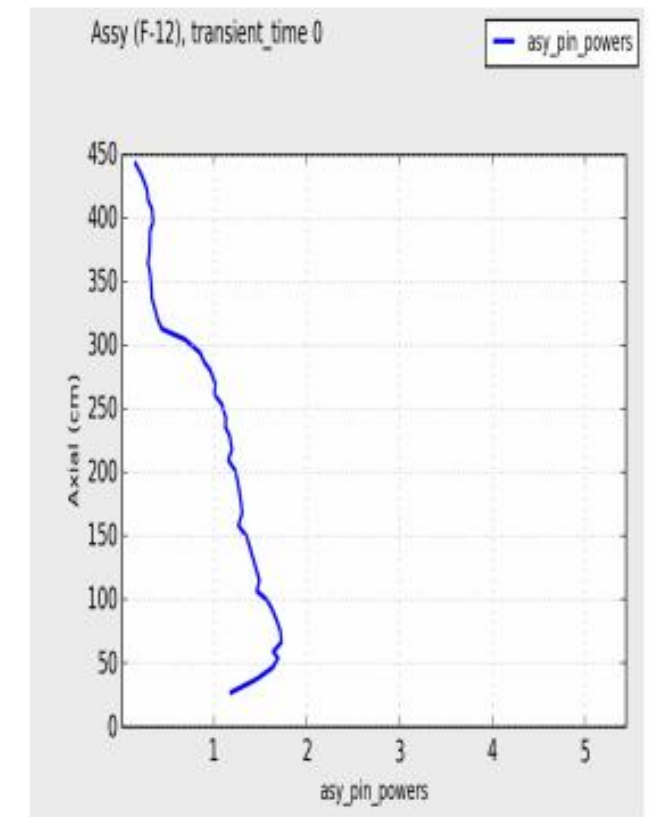
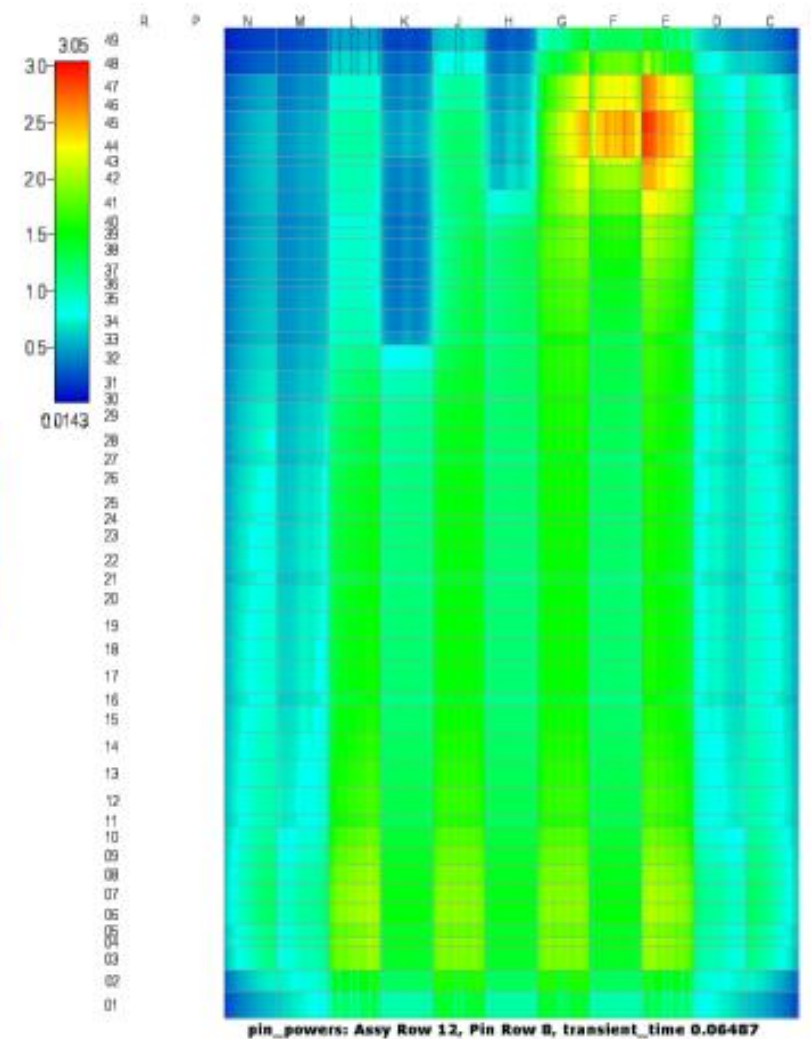
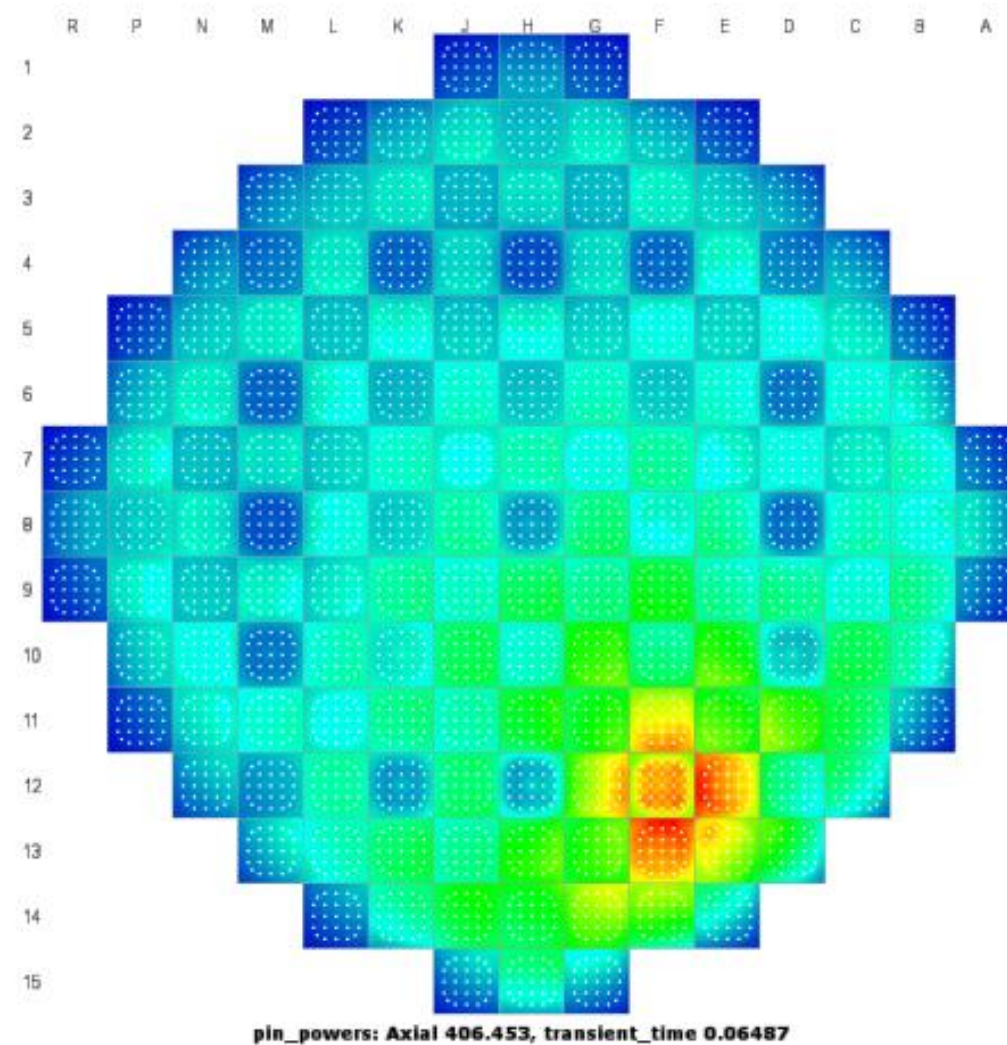


Multi-physics coupling and data exchange for high fidelity CRUD deposition simulations



Comparison between STAR-CCM+/MAMBA simulation (blue) and plant data (red dots) for selected fuel pin in Seabrook PWR fuel assembly G63

Example of VERA high fidelity capability

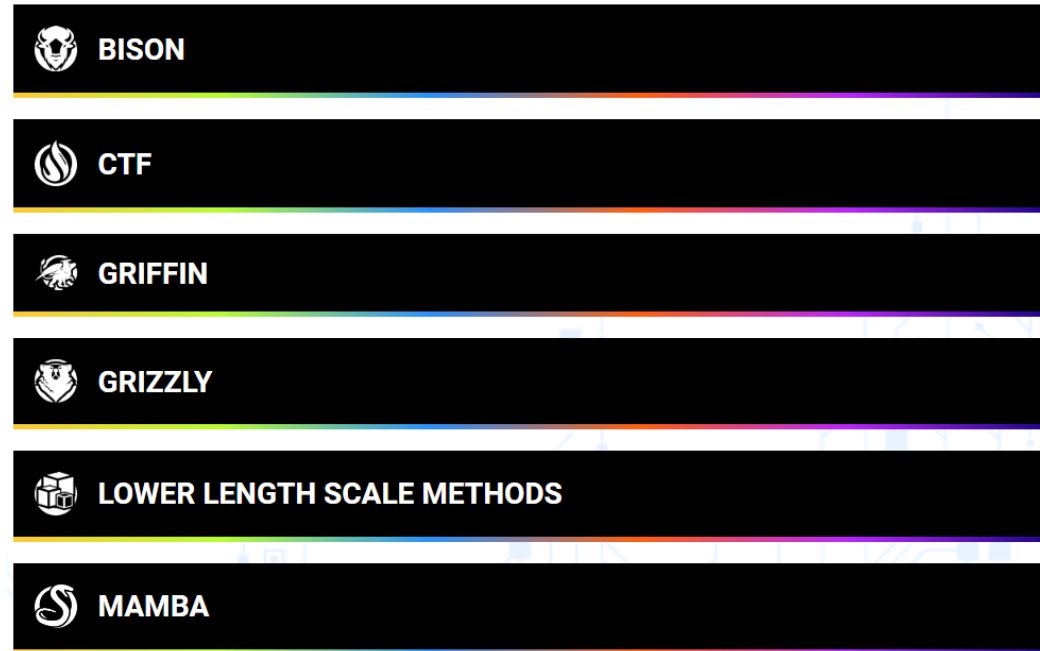


Highly skewed axial power initially with partially inserted rod

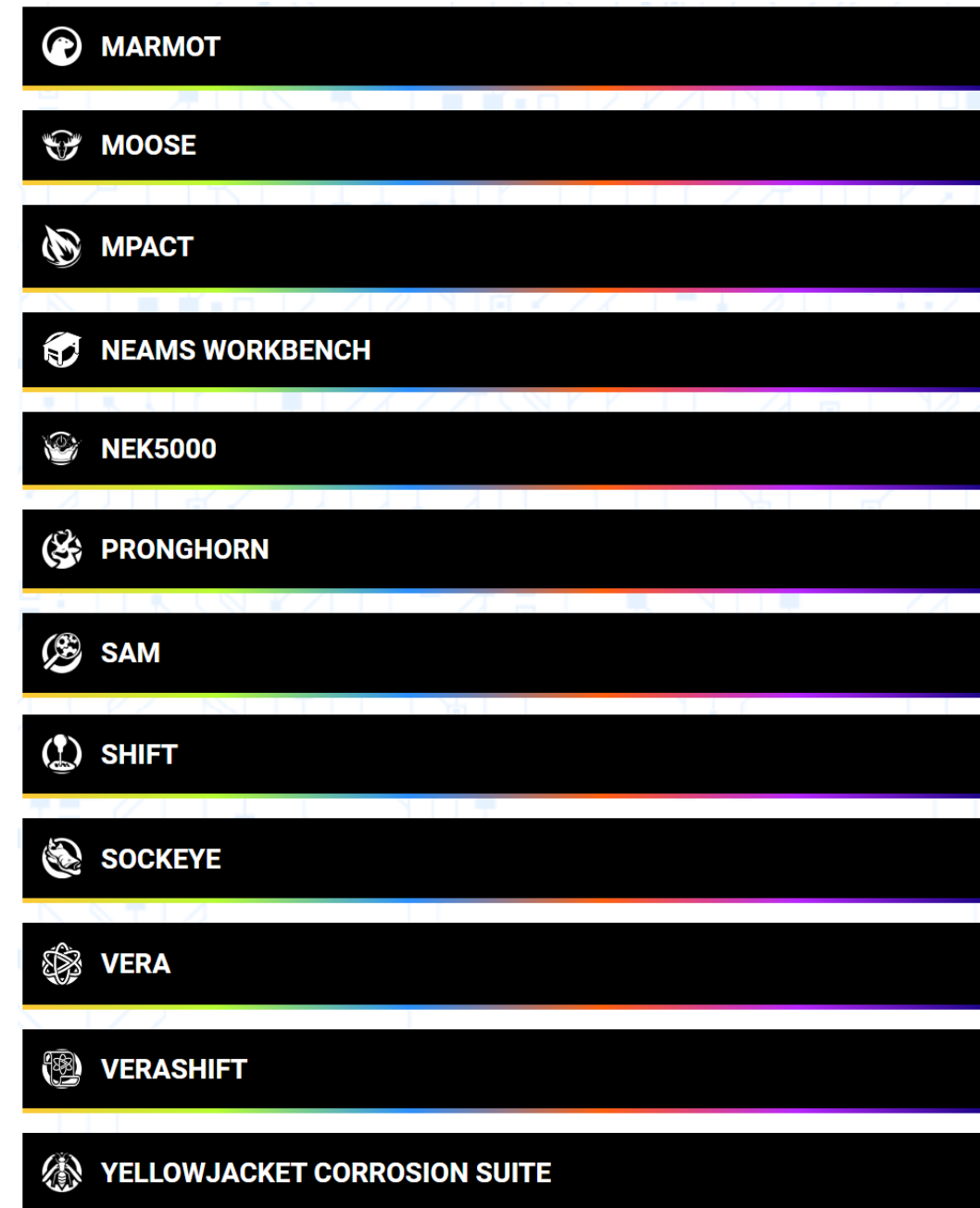
NEAMS: Nuclear Energy Advanced Modeling and Simulation DOE Program

- Program dedicated to the development, demonstration and deployment of advanced modeling and simulation tools and capabilities for:
 - General advanced nuclear energy technologies
 - LWRs
 - Non-LWRs
 - Advanced fuels
- NEAMS follows the three DOE-NE research objectives:
 1. Enhance the long-term viability and competitiveness of the existing U.S. reactor fleet LWRs
 2. Develop an advanced reactor pipeline
 3. Implement and maintain a national strategic fuel cycle and supply chain infrastructure

NEAMS: Nuclear Energy Advanced Modeling and Simulation DOE Program



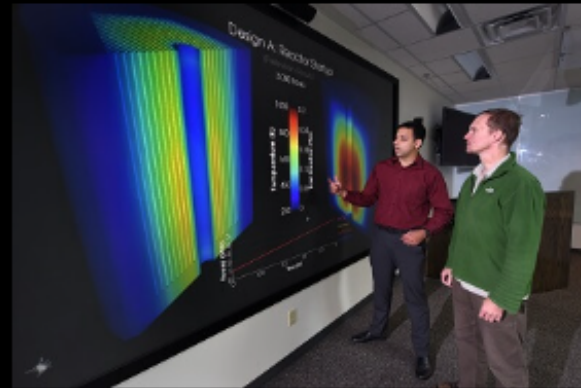
- You will already recognize some of the codes/platforms.
- Let's look briefly the codes/platforms we are not familiar with.



NEAMS: Nuclear Energy Advanced Modeling and Simulation DOE Program



MOOSE



Capabilities:

The foundation of the multiphysics simulations is based upon the MOOSE framework. The use of MOOSE as the library for multiphysics applications standardizes code inputs, communication, and linking. The MOOSE framework is well suited to solve a wide range of fully-coupled physics problems. Additionally, MOOSE as a platform gives developers access to a wide range of tools and processes to build production quality applications including the pluggable, parallel regression test system and NQA-1 compliant processes.

MOOSE is a finite-element framework that aids in application development by harnessing state-of-the-art fully-coupled, fully-implicit multiphysics solvers while providing automatic parallelization, mesh adaptivity, and a growing set of physics modules. It is the framework upon which several of the Nuclear Energy Advanced Modeling and Simulation (NEAMS) program codes are based, including BISON, SAM and Pronghorn. It has also been used to successfully couple MOOSE-based codes (such as BISON) to external codes such as Nek5000 and the NRC's FAST and TRACE applications.

Applications:

MOOSE is the underlying framework for most of the NEAMS physics tools as well as the capability for connecting multiple physics applications together. Individual codes employing MOOSE include Griffin, BISON, SAM, Pronghorn, and Grizzly.

Integrations:

MOOSE serves as the integrating tool for all applications built upon the framework as well as those that have been "wrapped" by MOOSE including Nek5000, MPACT, OpenMC, TRACE, and FAST

Coming Soon:

MOOSE recently gained an automatic differentiation capability and as well as finite volume discretization that can be coupled with existing finite element simulations. Improvements in dynamic application loading are being planned that will enable more flexible use of different combinations of applications for complex multiphysics simulations in support of advanced reactor concepts.

NEAMS: Nuclear Energy Advanced Modeling and Simulation DOE Program



GRIFFIN

Capabilities:

Developed in FY 2020, Griffin is a time-dependent finite-element based reactor physics code built using the MOOSE framework with weak form formulations for diffusion, PN, and 1st and 2nd-order SN transport and a variety of equivalence techniques

Applications:

Griffin is specially designed to support multi-physics applications and to natively couple to any MOOSE application. It supports both two-step and on-line cross sections preparation and includes state-of-the-art depletion solvers.

Integrations:

Griffin integrates two previously code suites, the INL-developed MAMMOTH/Rattlesnake code suite and the ANL-Developed MC2-3/PROTEUS code suite

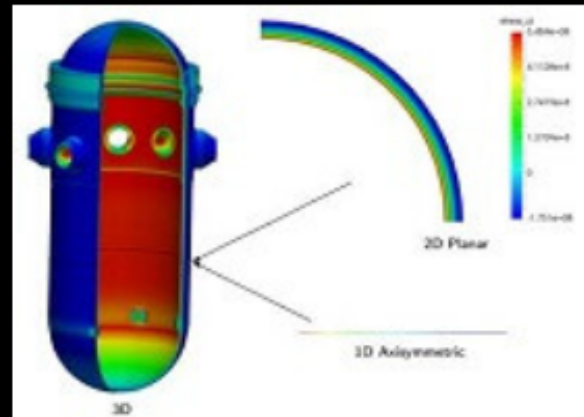
Coming Soon:

Improved heterogeneous transport and on-line cross section preparation

NEAMS: Nuclear Energy Advanced Modeling and Simulation DOE Program



GRIZZLY



Grizzly calculation of thermo-mechanical response of a light water reactor (LWR) reactor pressure vessel (RPV) under transient loading conditions.

Capabilities:

Grizzly simulates progression of aging mechanisms in nuclear power plant components and the integrity of these components subjected to degradation during normal and off-normal loading conditions. BlackBear is an open-source code, with non-nuclear specific models, that provides a subset of Grizzly's capabilities.

Applications:

- LWR Reactor pressure vessel (RPV) probabilistic fracture mechanics (PFM): 1D, 2D, or 3D modeling of RPV thermo-mechanical response during pressurized thermal shock (PTS) transient events. Semi-empirical model currently used for embrittlement prediction. PFM model evaluates the probability of large sets of flaws using Monte Carlo analysis.
- Concrete degradation: Coupled physics (thermal and moisture transport and mechanics) modeling of processes leading to expansive reactions in concrete due to alkali-silica reaction (ASR) and radiation induced volumetric expansion (RIVE)
- High temperature creep modeling: Models for high temperature creep of advanced reactor structural materials using both traditional phenomenological approach and a reduced order model based on material microstructure

Integrations:

- VERAShift: Provides 3D fluence maps used for radiation-induced damage in RPVs
- NEML: Material constitutive model library that provides phenomenological high temperature creep models
- MASTODON: MOOSE-based code for simulation of seismic response of structures

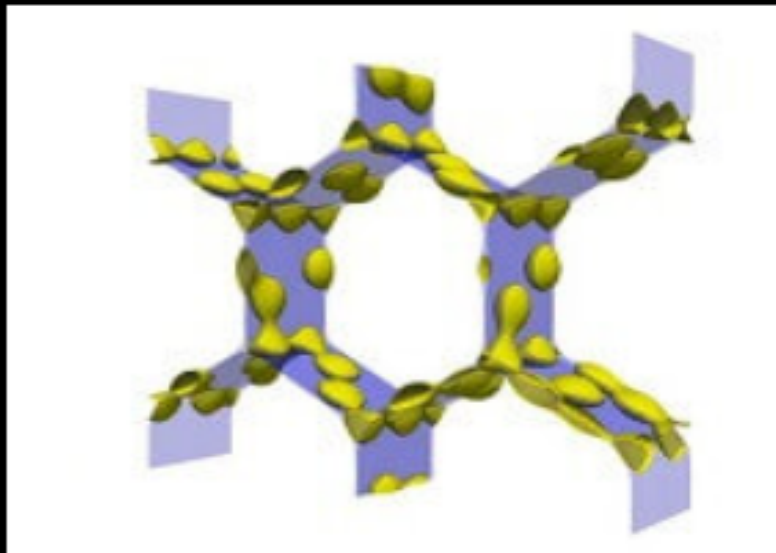
Coming Soon:

- Coupling with 3D computational fluid dynamics (Nek5000) to provide thermal boundary conditions for RPV during PTS
- Predictive models for LWR RPV steel embrittlement based on multiscale modeling are under development

NEAMS: Nuclear Energy Advanced Modeling and Simulation DOE Program



MARMOT



Simulation of fission gas bubble growth along grain boundaries and triple junctions in UO₂ nuclear fuel during operation.

Capabilities:

Marmot solves the phase-field equations that resolve evolving microstructure features (mesoscale) and is used to develop empirical-like material models used at the engineering scale (e.g. BISON)

Applications:

Supports all fuel types in BISON

Integrations:

Marmot can be coupled to the binary collision Monte Carlo radiation damage application, Magpie, and engineering scale codes such as BISON or Grizzly

Coming Soon:

Coupling to commercial thermodynamic and kinetic databases for high fidelity materials models

NEAMS: Nuclear Energy Advanced Modeling and Simulation DOE Program

**PRONGHORN**

Molten salt reactor modeling.

Capabilities:

Pronghorn is a multi-dimensional, coarse-mesh, thermal-hydraulics (TH) code for advanced reactors. It serves the intermediate fidelity realm situated between detailed computational fluid dynamics (CFD) analysis and lumped system models.

Applications:

- Pronghorn is particularly well-suited for gas-cooled pebble-bed (PBR) and prismatic reactors, fluoride-salt cooled high-temperature reactors (FHR), and open-pool molten salt reactors (MSRs).
- The main area of application for Pronghorn is the analysis of accident scenarios of advanced reactors, e.g. loss of flow and overcooling (among others). Pronghorn's coarse mesh approach also enables rapid scoping studies and design iterations.

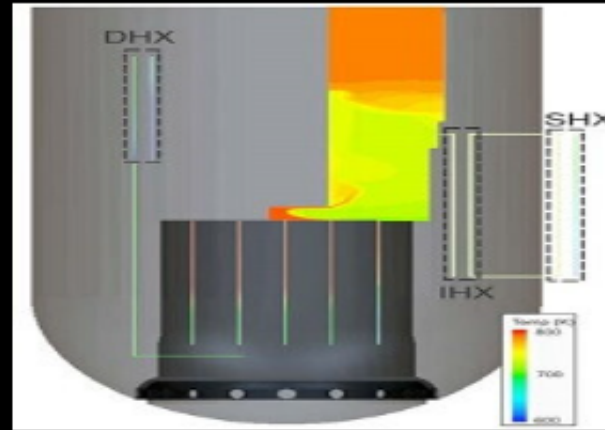
Integrations:

- Pronghorn is built and tested with the newest version of MOOSE and follows the MOOSE development paradigm. Therefore, Pronghorn leverages the seamless multiapplication data transfer and execution capabilities of MOOSE and enables thermalhydraulic feedback in many other MOOSE applications.
- Pronghorn readily integrates with the Thermal-Hydraulics Module (THM) for coupled core/primary loop/secondary loop TH analysis and prediction of bypass flows.
- Pronghorn has been coupled to Griffin deterministic neutronics for a number of applications, such as coupled steady-state, 100-40-100 load follow, and LOFC transient for the PBMR-400 benchmark and precursor drift feedback for pool-type MSRs.
- A multiscale coupling of macro, meso, and micro length scale models based on MOOSE's multi-application system is used to predict pebble bed reactor TRISO transient thermal performance.

Coming Soon:

- Pronghorn currently uses a continuous finite element method (CFEM) discretization. For more robust solution, a finite volume discretization is being added with both compressible and Boussinesq-type incompressible formulations.
- Turbulence modeling
- Coupling with NEAMS system analysis tools such as THM and SAM to efficiently model 1-D components outside of the multi-dimensional core region
- Advanced plug-and-play interface for agnostic porous media calculations for importing custom closures from higher fidelity calculations. This capability will allow computationally efficient analysis of complex phenomena such reflector bypass flows and the near-wall physics in PBRs.
- Improvement and validation of correlations for pebble bed reactors using detailed, spatiotemporally resolved data.

NEAMS: Nuclear Energy Advanced Modeling and Simulation DOE Program



Temperature distribution in the coupled SAM-CFD simulations of advanced burner test reactor protected loss-of-flow transient.

Capabilities:

System Analysis Module (SAM) is a fast-running, whole-plant transient analysis code with improved-fidelity capability for fast turnaround design scoping and safety analyses of advanced non-light-water reactors.

- One-D flow networks represent general fluid systems such as the reactor coolant loops
- Multi-channel representation of fuel assembly and inter-assembly heat transfer modeling
- Point kinetics model with various reactivity feedback including thermal expansions
- Specific component models for reactor transient simulations
- A general mass transport capability for species transport
- Control and trip system models
- Reduced-order multi-dimensional flow models for thermal mixing and stratification modeling

Applications:

System transient analysis of various advanced reactor types for design optimization and licensing support, under operational transients and accident conditions

Integrations:

Coupling with Mammoth, Proteus, Nek5000, and BISON, STAR-CCM+, SAS4A/SASSYS-1, and TRACE

Coming Soon:

- Fluid freezing/thawing modeling capability for modeling of overcooling events in liquidsalt-cooled reactors
- General mass transport modeling capability in the reduced-order multi-dimensional flow model, for application to molten salt fueled (fast spectrum) reactors
- Tritium transport modeling in molten salt cooled reactors

NEAMS: Nuclear Energy Advanced Modeling and Simulation DOE Program

**SOCKEYE**

Capabilities:

Sockeye is a MOOSE-based heat pipe simulator and analysis tool. Heat pipe simulation offers the ability to accurately predict heat transfer for applications involving heat pipes, including heat-pipe-cooled microreactors. Additionally, it provides insight on heat pipe performance; operational heat pipe limits can be predicted in transient conditions and with greater flexibility than steady-state analyses can provide. Heat pipe simulation can be used by industry to inform heat pipe and microreactor design. Simulations can be performed to determine the range of operating conditions applied to a heat pipe and whether reliable heat pipe operation can be sustained in these conditions. If operational limits are encountered, heat pipe design can be altered to avoid these limits. The following is a summary of Sockeye's capabilities:

- Transient, 1-D, two-phase flow coupled to 2-D heat conduction
- Transient, 2-D, effective thermal conductivity model
- Analytic tools for analyzing heat pipe operating envelope

Applications:

Heat pipe transient simulation

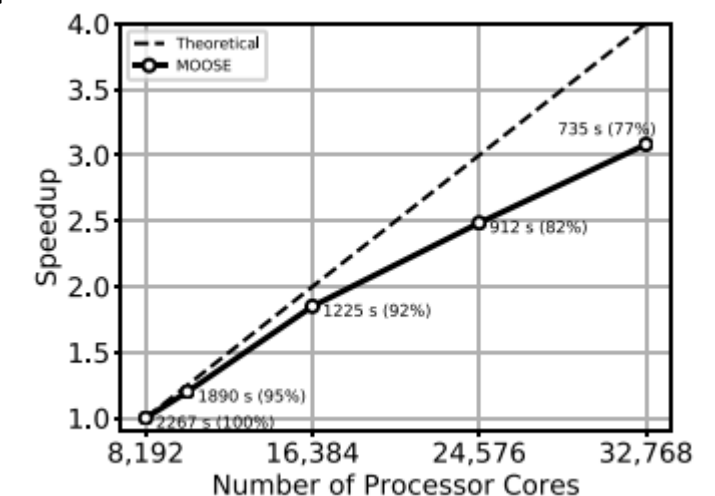
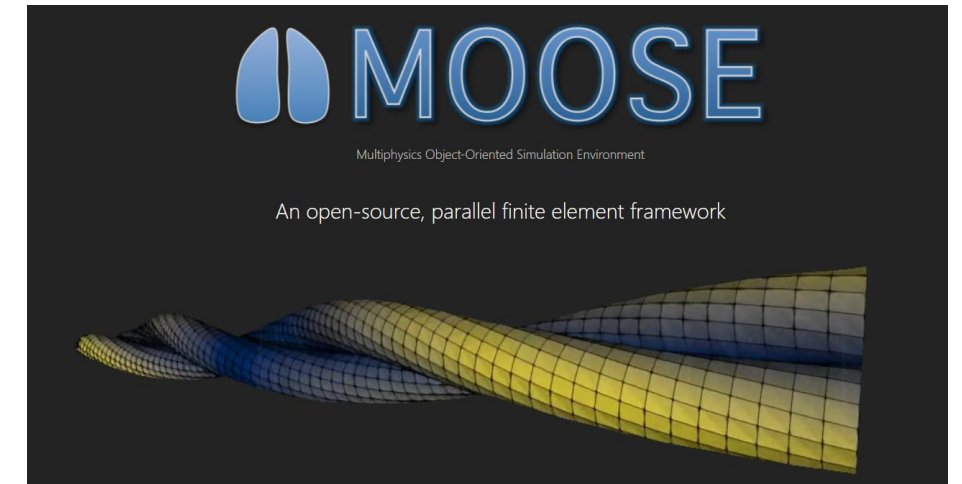
Integrations:

Sockeye couples via heat transfer to other physics applications in the DireWolf microreactor simulator application

Multi-Physics Object Oriented Simulation Environment (**MOOSE**)

A finite element framework for the development of tightly coupled multi-physics solvers

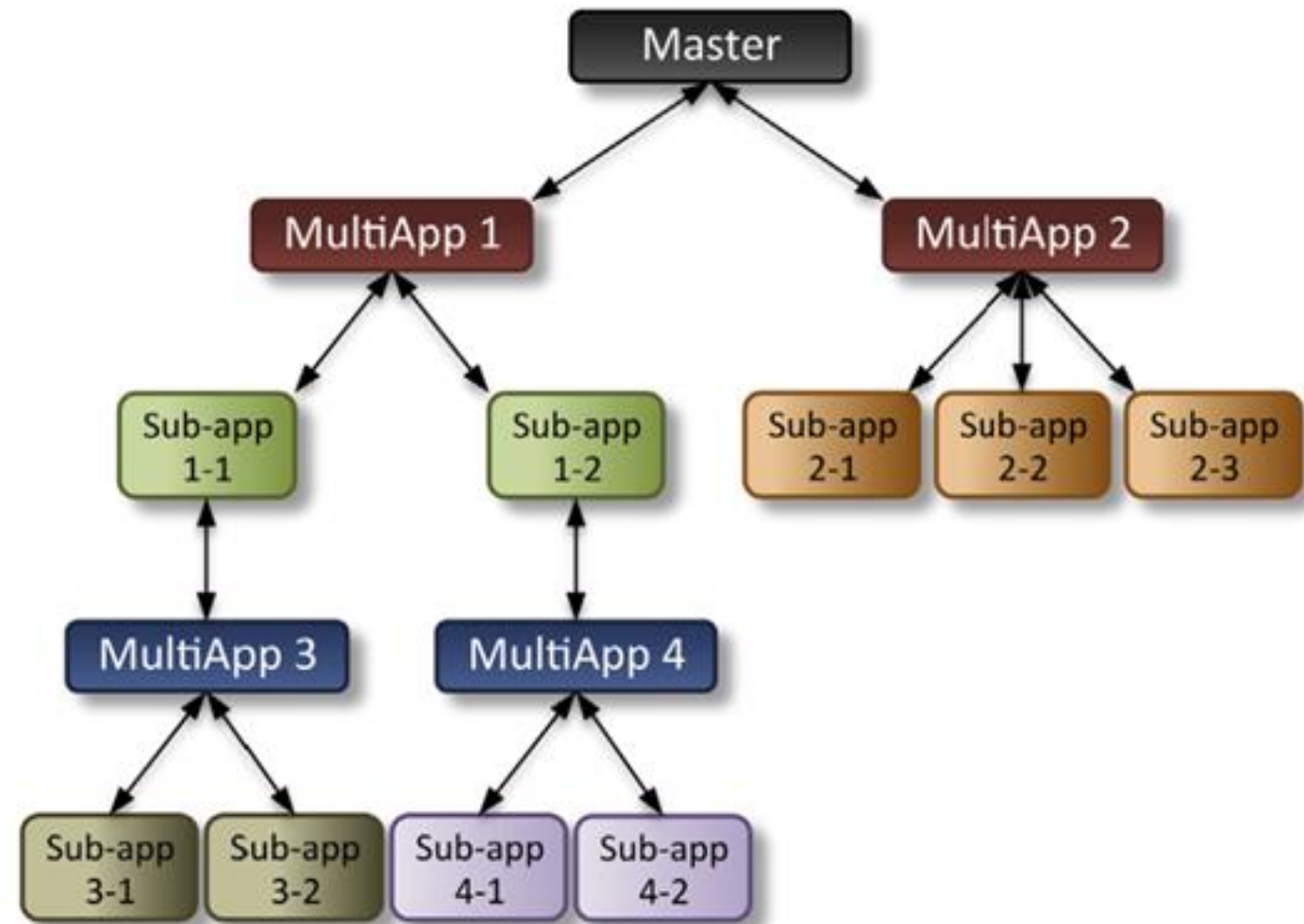
- Open-source general computational framework written in C++ initiated at INL around 2008.
- Numerical solutions using finite elements.
- Modular kernels are used that describe numerical operators (e.g. 1D-convection, 2D-diffusion).
- Various numerical PDEs can be solved by combining different kernels and boundary conditions.
- Jacobian-Free Newton Krylov (JFNK) method is used to numerically solve the PDFs
- Built-in tools for mesh generation, data transfer, and postprocessing.
- Implicit multi-physics non-linear solutions.
- Automatic differentiation.
- Parallel high performance computing capabilities.
- Many Codes/Applications have been developed using all these capabilities.



Multi-Physics Object Oriented Simulation Environment (**MOOSE**)

A finite element framework for the development of tightly coupled multi-physics solvers

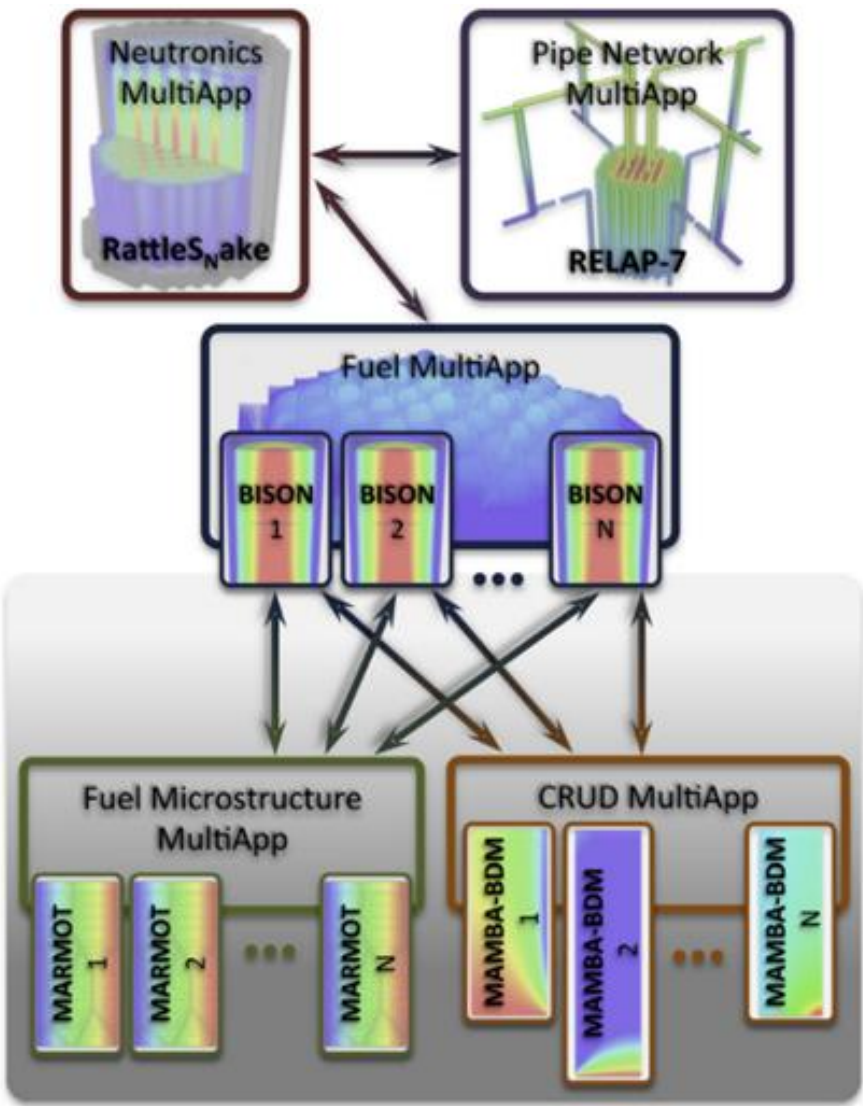
- A flexible MultiApp system that allows:
 - Individual developers to focus on their respective modeling applications.
 - If a multi-physics coupling is needed the applications can be efficiently coupled in a hierarchical structure without the need for developing a specific interface.
 - Applications outside of MOOSE can be also coupled by developing appropriate wrappers and interfaces.



Multi-Physics Object Oriented Simulation Environment (MOOSE)

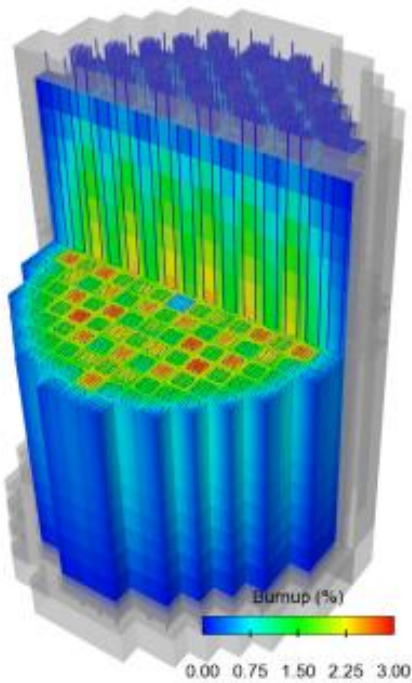
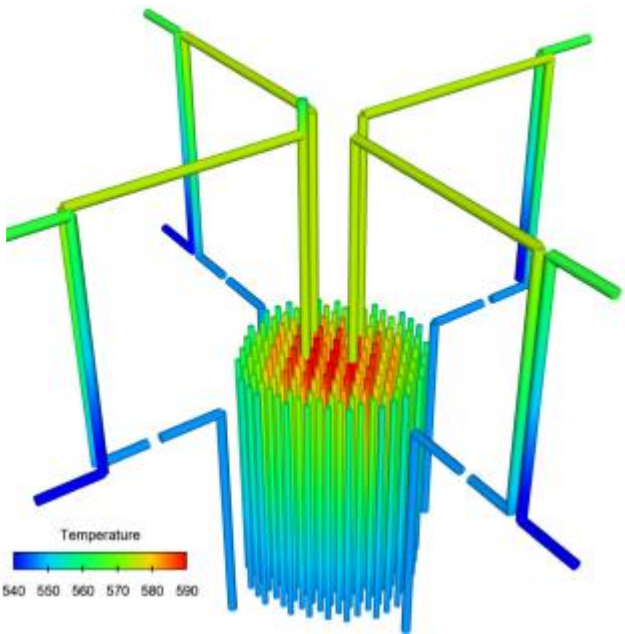
Simulation data flow chart for RattleSnake, RELAP-7, and BISON

Microstructure calculations conducted with the MARMOT and MAMBA-BDM applications (shaded region) were also used in some part of the simulation



Application	SubApp	Data
RattleSnake	→ BISON	Coolant Temp. Fission Rate
RattleSnake	← BISON	Fuel Temp. Clad Temp. Burnup
RattleSnake	→ RELAP-7	Clad Temp.
RattleSnake	← RELAP-7	Coolant Temp.
BISON	→ MARMOT	Fuel Temp. Fission Rate
BISON	← MARMOT	Thermal Cond.
BISON	→ MAMBA-BDM	Clad Temp. Coolant Temp.
BISON	← MAMBA-BDM	BO ₃ Conc. CRUD Temp.

Details of Transfers used in the simulations for RattleSnake, RELAP-7, and BISON used in the calculations

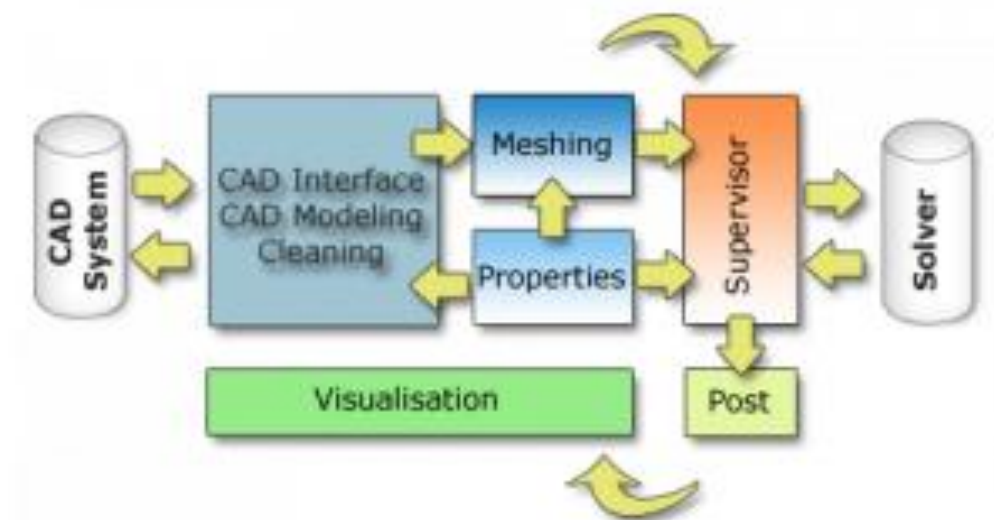


Simulation numérique par Architecture Logicielle en Open source et à Méthodologie d'Évolution (**SALOME**)

French for “Numerical Simulation by Computing Architecture in Open Source and with Evolving Methodology”

- Open-source general computational framework written in C++ and Python, developed by EDF/CEA in France since 2000.
- Compared to MOOSE there is more focus on meshing, data transfer, visualization and postprocessing than modular numerical solution of PDEs.
- The platform is flexible and allows the development of numerical applications similar to the MultiApp system in MOOSE.
- Its main goal is to provide a framework for code-agnostic multi-physics coupling.
- It includes frameworks for data assimilation and uncertainty analysis.

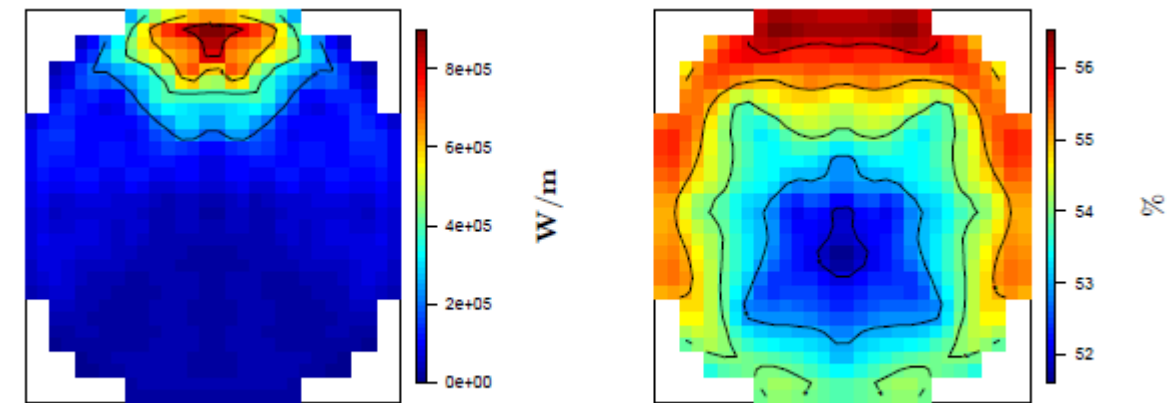
SALOME



Simulation numérique par Architecture Logicielle en Open source et à Méthodologie d'Évolution (**SALOME**)

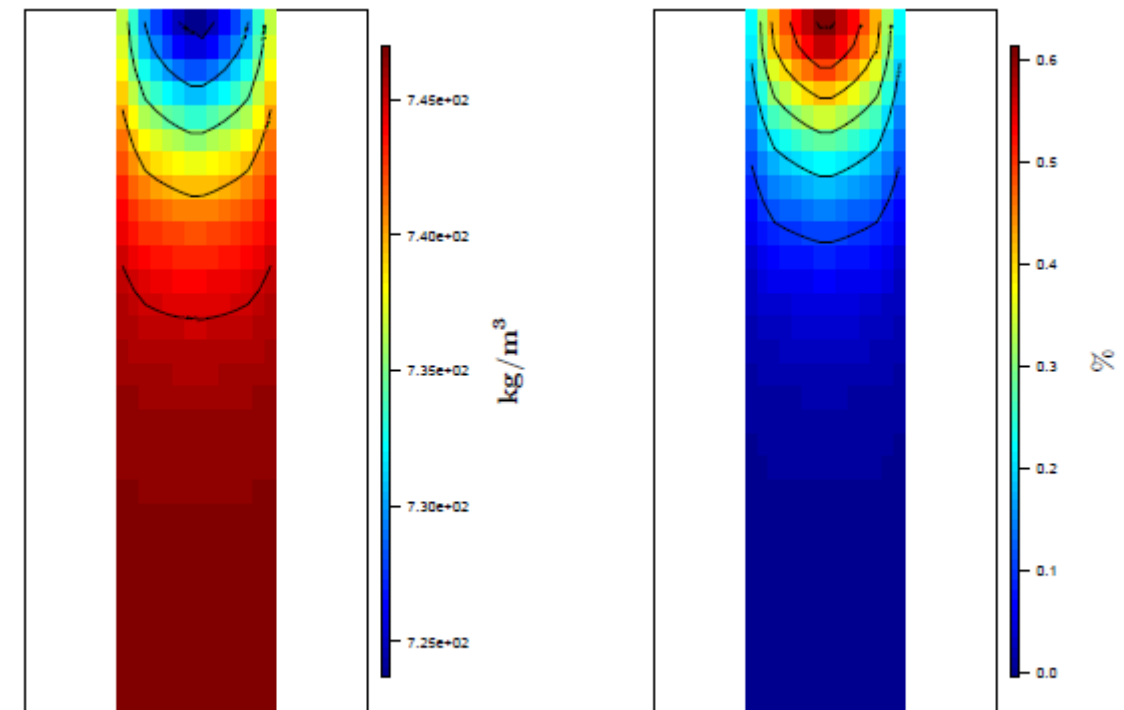
CORPUS/C3PO is an example of an application based on SALOME and developed by CEA

- CORPUS is multi-physics LWR application.
- APOLLO3 is a finite elements neutronics code suite that includes many solvers (e.g. MOC, Diffusion, Sn, SPn)
- FLICA4 is a 3D two-phase flow porous medium thermal-hydraulics code.
- ALCYONE is a 3D finite elements fuel performance code.
- The three codes are coupled to perform a rod ejection accident.
- Uncertainty analysis is performed including a Hi2Lo model for fuel performance.



(a) Mean

(b) Relative standard deviation



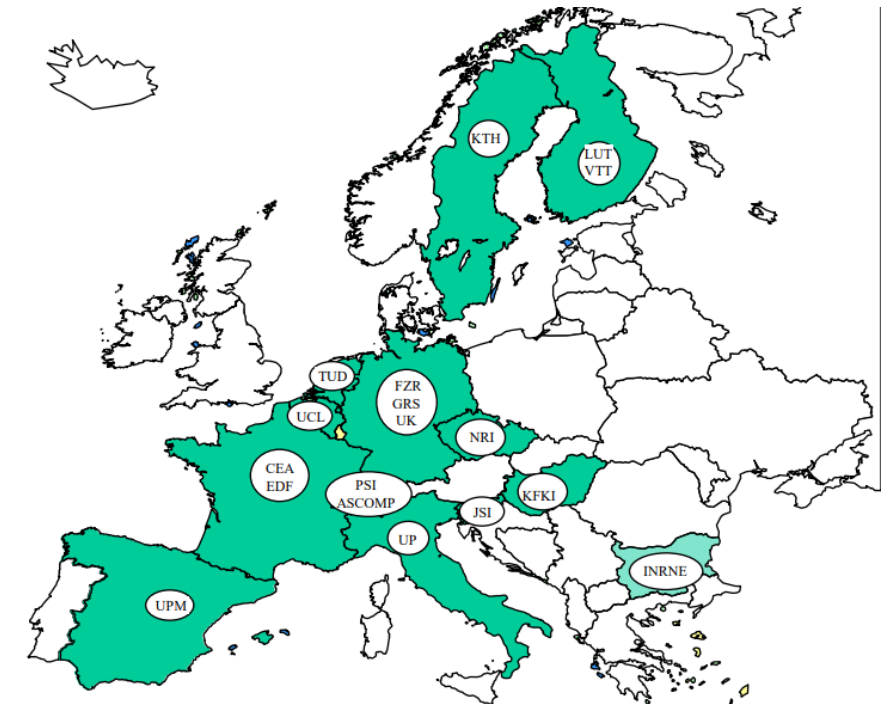
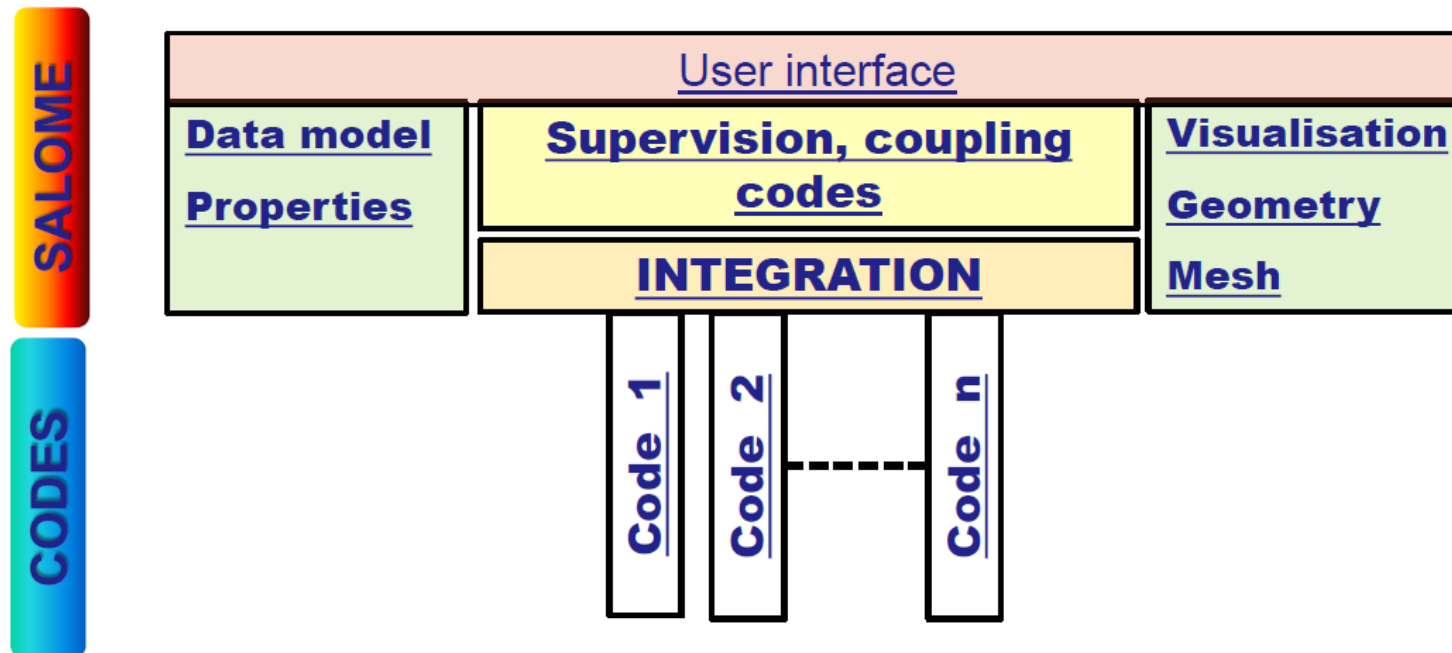
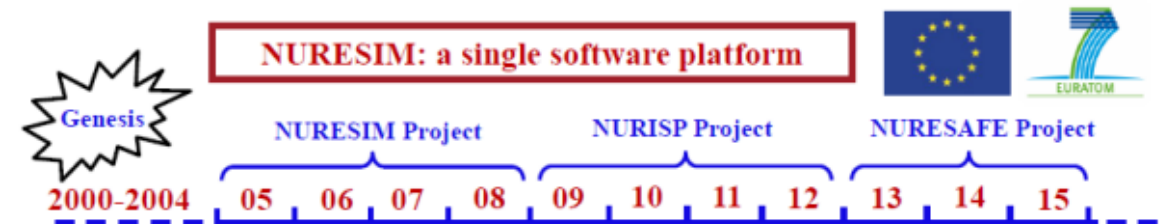
(a) Mean

(b) Relative standard deviation

NURESAFE/NURISP/NURESIM

European Project to Develop a Reference Simulation Platform for Nuclear Reactors

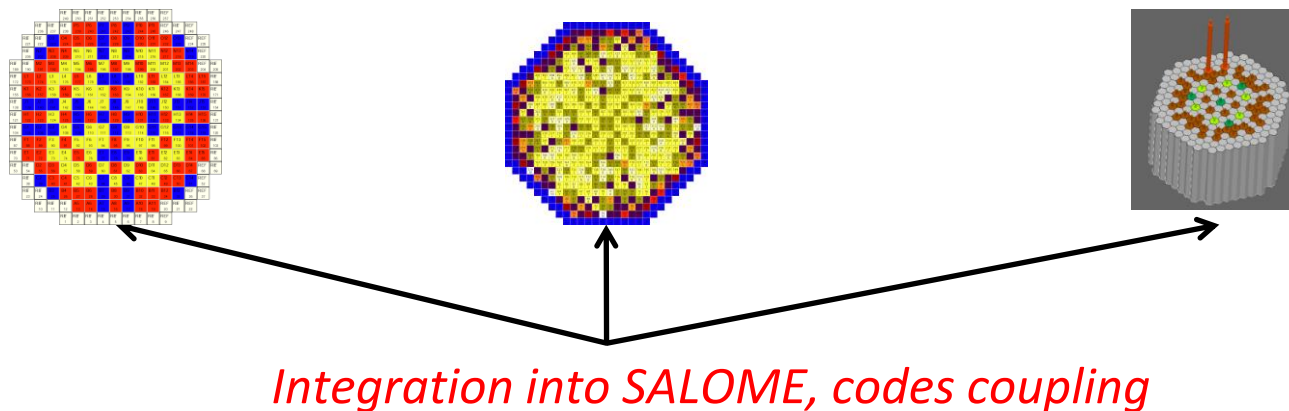
- International collaboration within the European Union to develop an integrated multi-physics and multi-scale platform for LWR and Non-LWRs initiated around 2005.
- Emphasis on VVUQ.
- It leverages SALOME capabilities.



European Project to Develop a Reference Simulation Platform for Nuclear Reactors

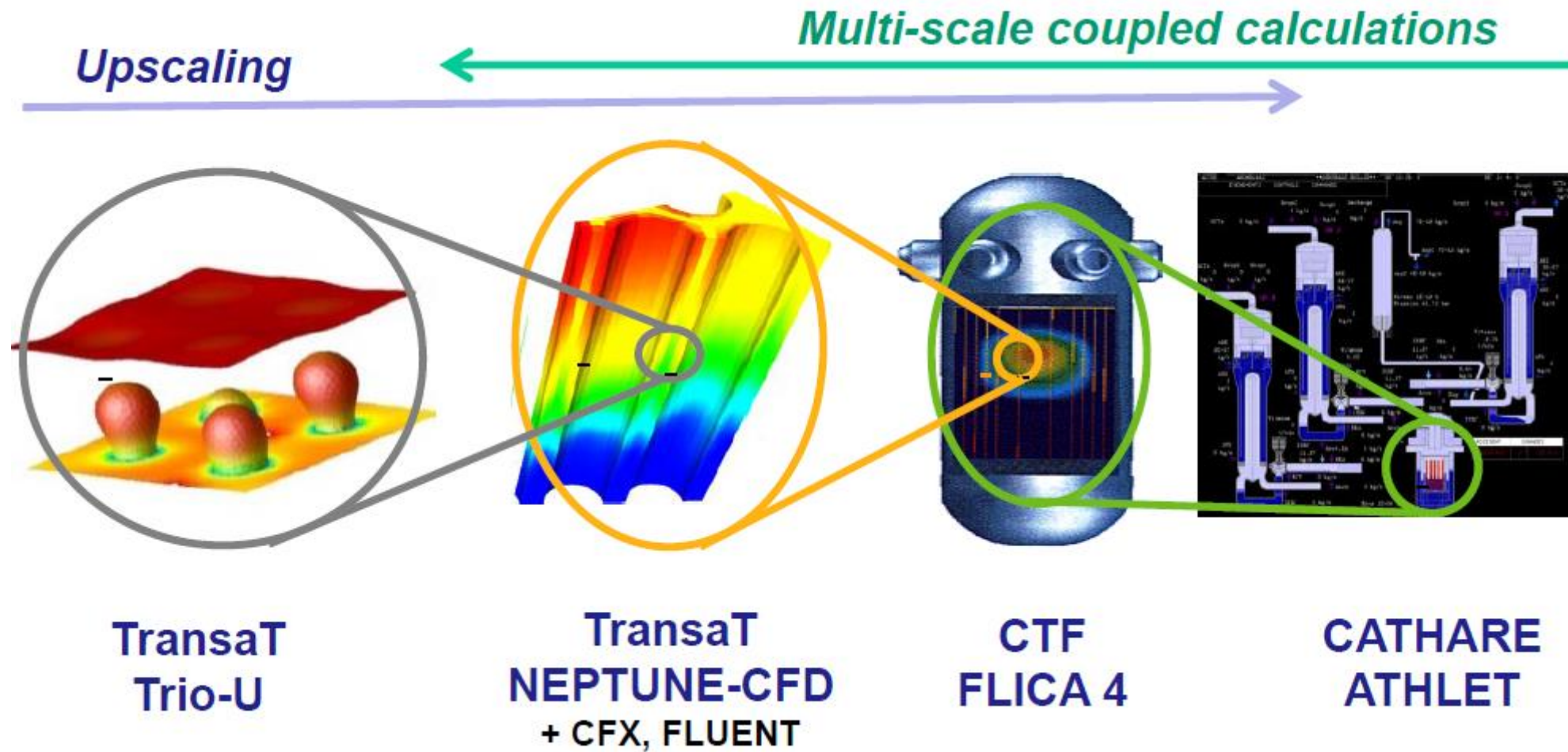
- ✓ Develop and test enhanced and new coupling schemes, conduct transient simulations at higher fidelity
- ✓ Simulation of three LWR transients:

VVER: MSLB



NURESAFE/NURISP/NURESIM

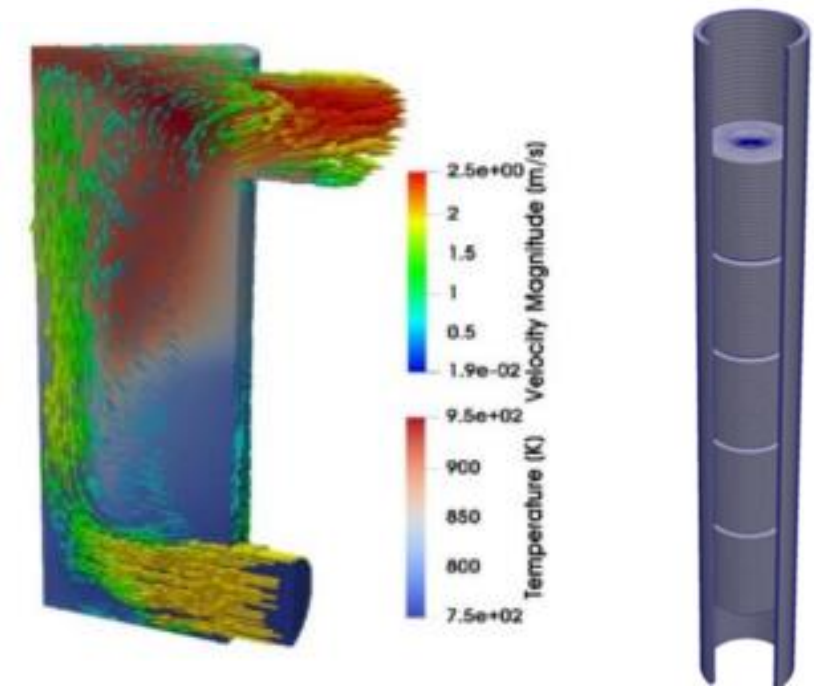
European Project to Develop a Reference Simulation Platform for Nuclear Reactors



Open Field Operation And Manipulation(OpenFOAM)

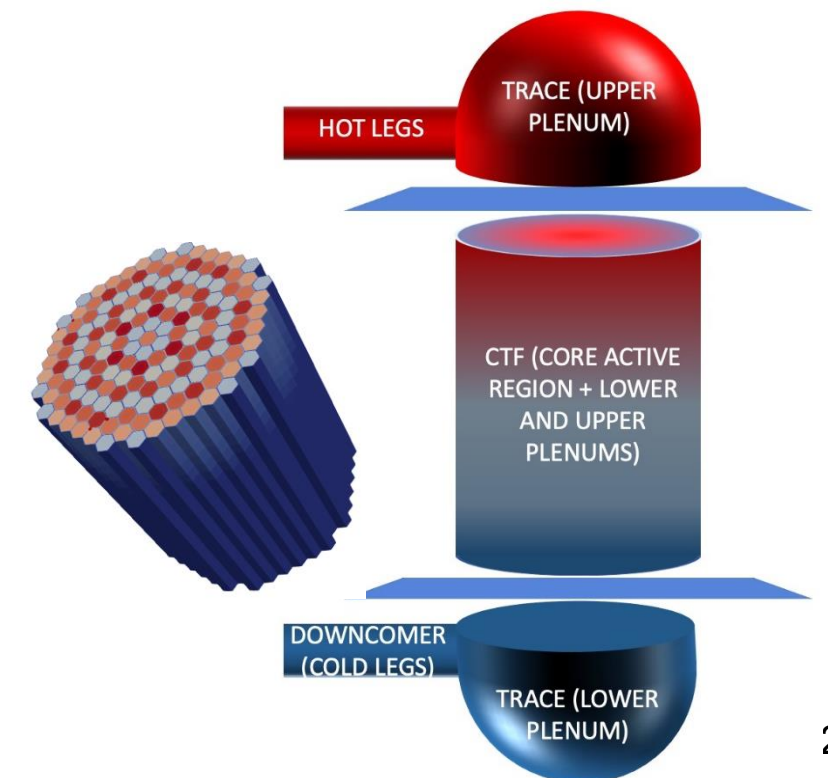
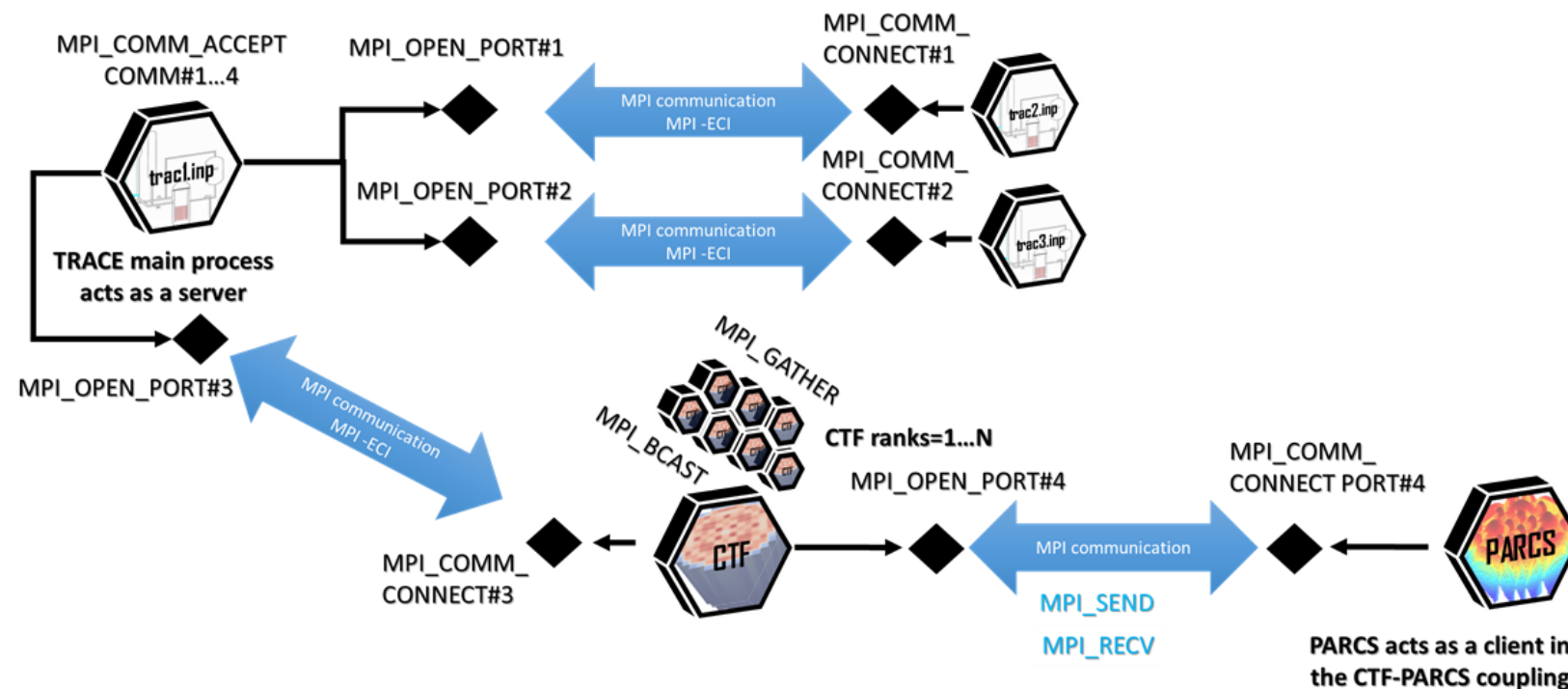
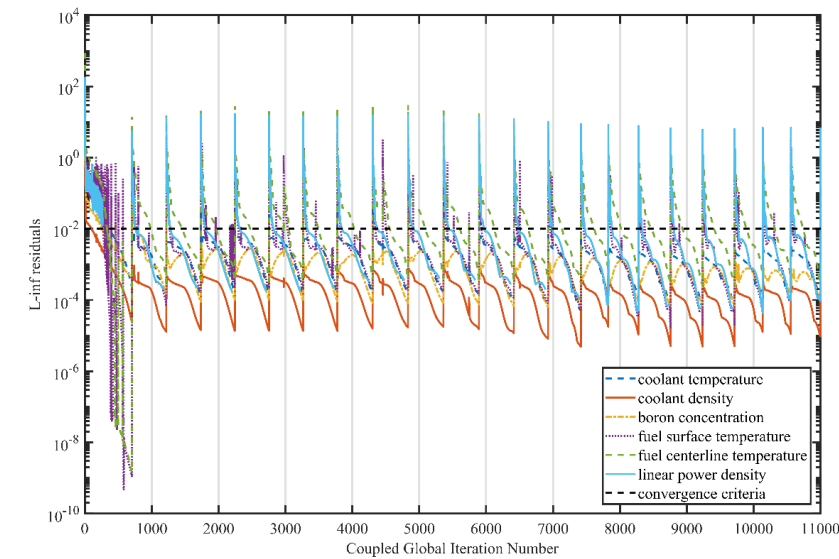
A finite volume framework for the development of numerical solvers

- Open-source general computational toolbox written in C++ for the flexible development of numerical solvers developed since 2004 in England.
- It contains libraries for efficient discretization, and solution of PDEs using finite volume methods.
- It contains various thermal-hydraulics and thermal-mechanics solvers.
- It is primarily used for CFD calculations.
- Recently, nuclear engineering applications have been developed with the more prominent ones being:
 1. Gen-FOAM (transient multi-physics analysis of advanced reactors)
 2. OFFBEAT (fuel performance)



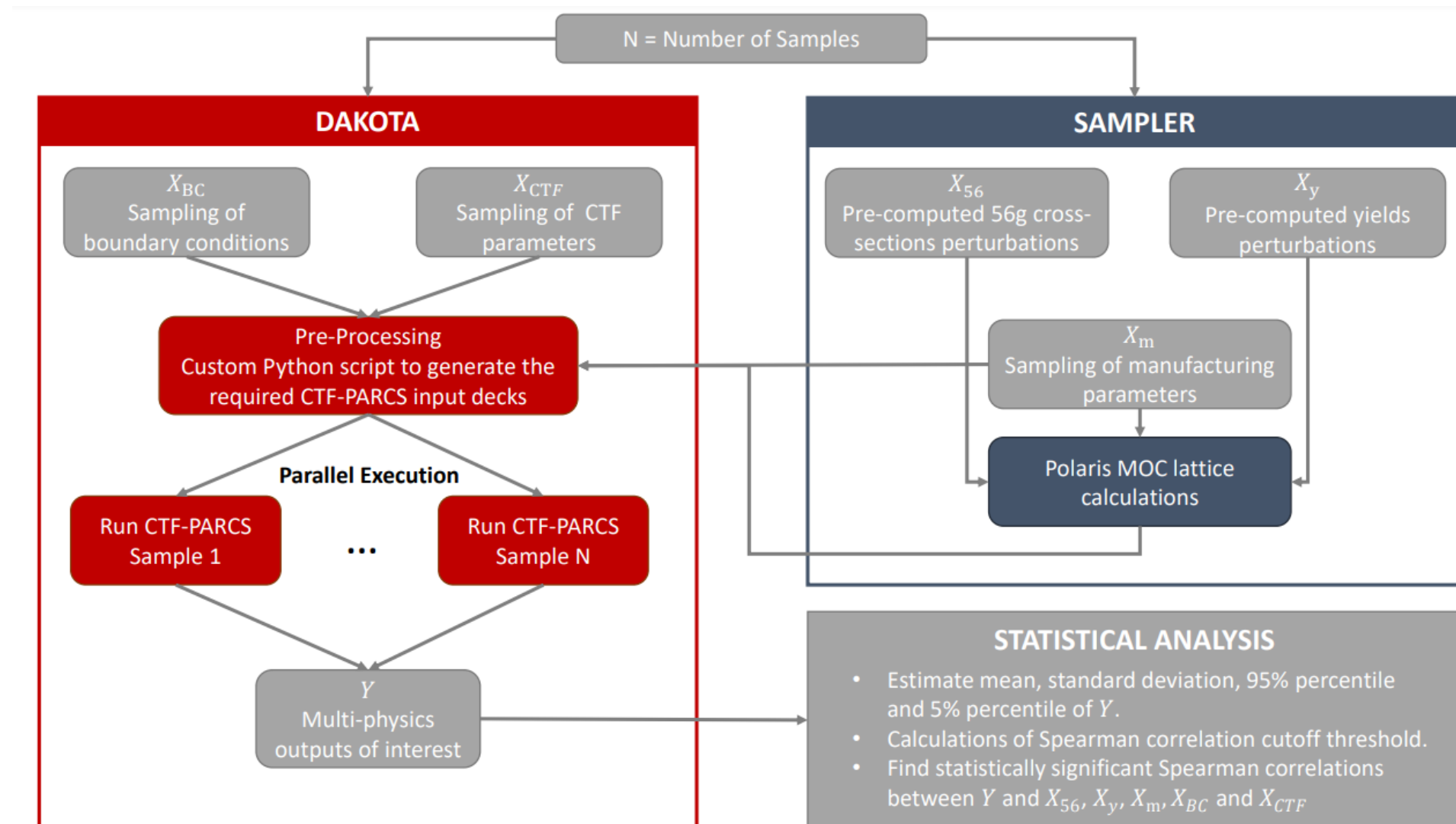
NCSU MULTI-PHYSICS PLATFORM FOR UNCERTAINTY QUANTIFICATION

- Traditional means improved industry/regulation-grade codes for LWRs
- Full-core channel- or subchannel-resolved thermal-hydraulics model with CTF (3D channel or subchannel models).
- Full-core assembly-resolved neutronics model with PARCS (3D diffusion) with full core pin-power reconstruction.
- CTF-PARCS coupled code for the multi-physics modeling of the core and TRACE for the system thermal-hydraulics.
- The coupling covers: steady state, depletion and transient conditions.



NCSU MULTI-PHYSICS PLATFORM FOR UNCERTAINTY QUANTIFICATION

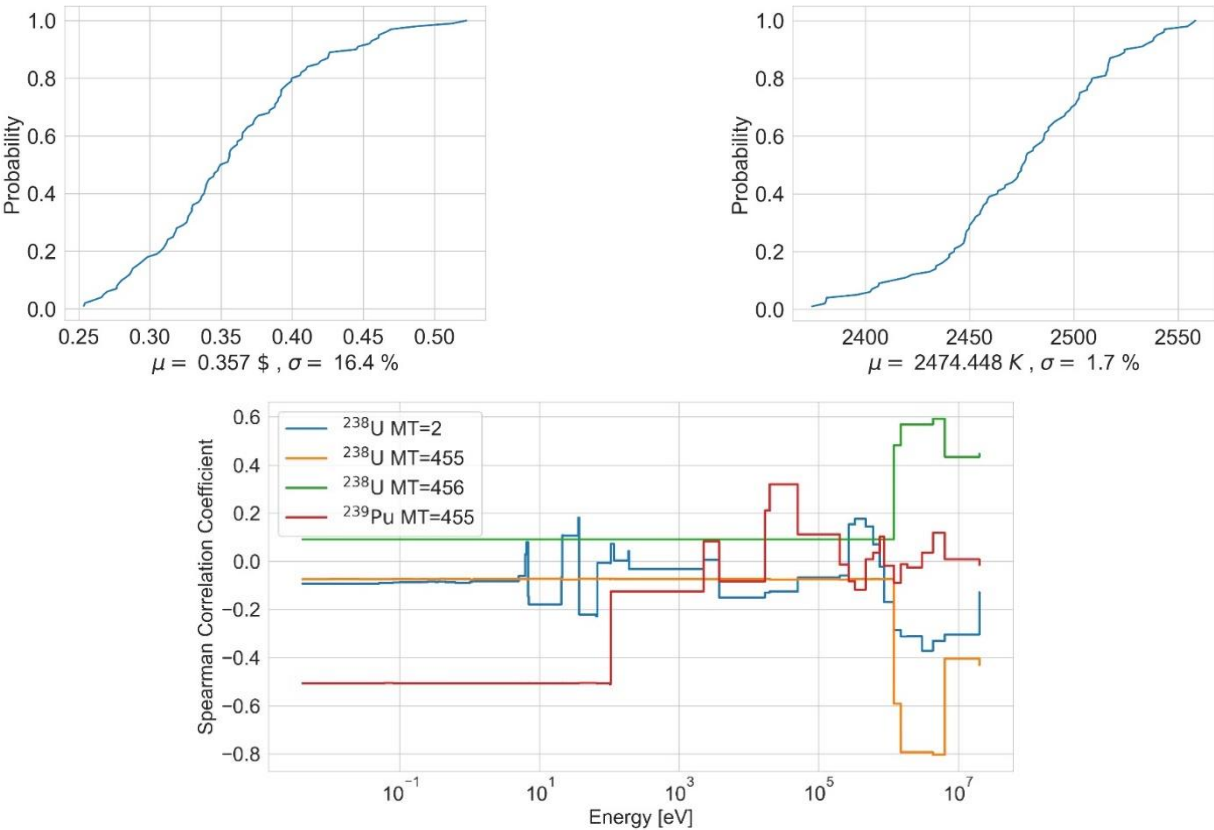
- UQ framework consisting of a combination of case dependent and case independent perturbations.
- The case independent perturbations are pre-generated and stored.



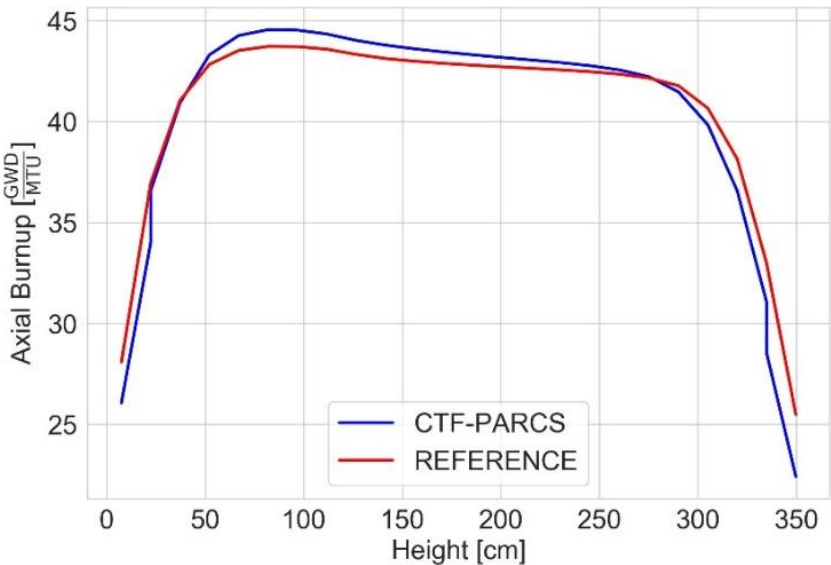
NCSU MULTI-PHYSICS PLATFORM FOR UNCERTAINTY QUANTIFICATION

- Testing for PWR - Application to LWR-UAM Phase III.
- PWR TMI-1 core of 177 fuel assemblies and 1 ring of reflectors.
- Steady state, Depletion and Rod Ejection Accident (REA) transient cases.
- Hot Full Power (HFP), Hot Zero Power (HZP), Beginning of Cycle (BOC) and End of Cycle (EOC) conditions.

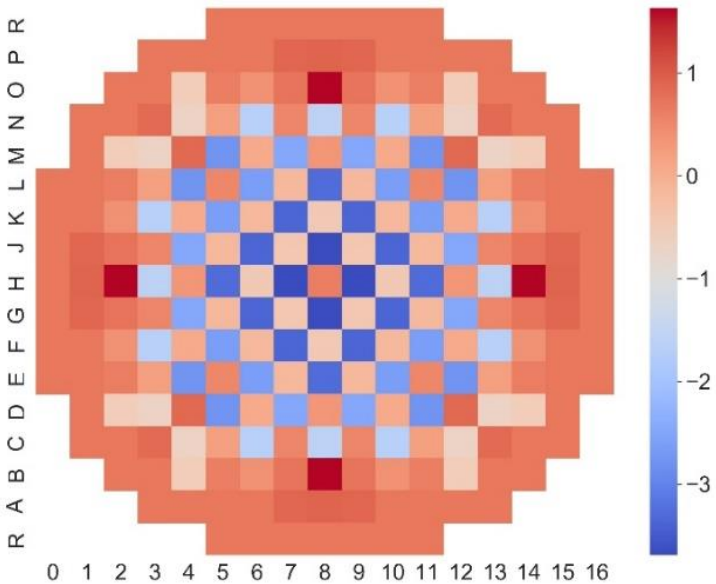
EOC REA Injected Reactivity and Maximum Temperature



EOC burnup axial profile



EOC burnup radial discrepancies



NCSU MULTI-PHYSICS PLATFORM FOR UNCERTAINTY QUANTIFICATION

- UQ framework consisting of a combination of case dependent and case independent perturbations.
- Similar to LWR framework but for HTGRs.
- VSOP code is used to model HTGRs.

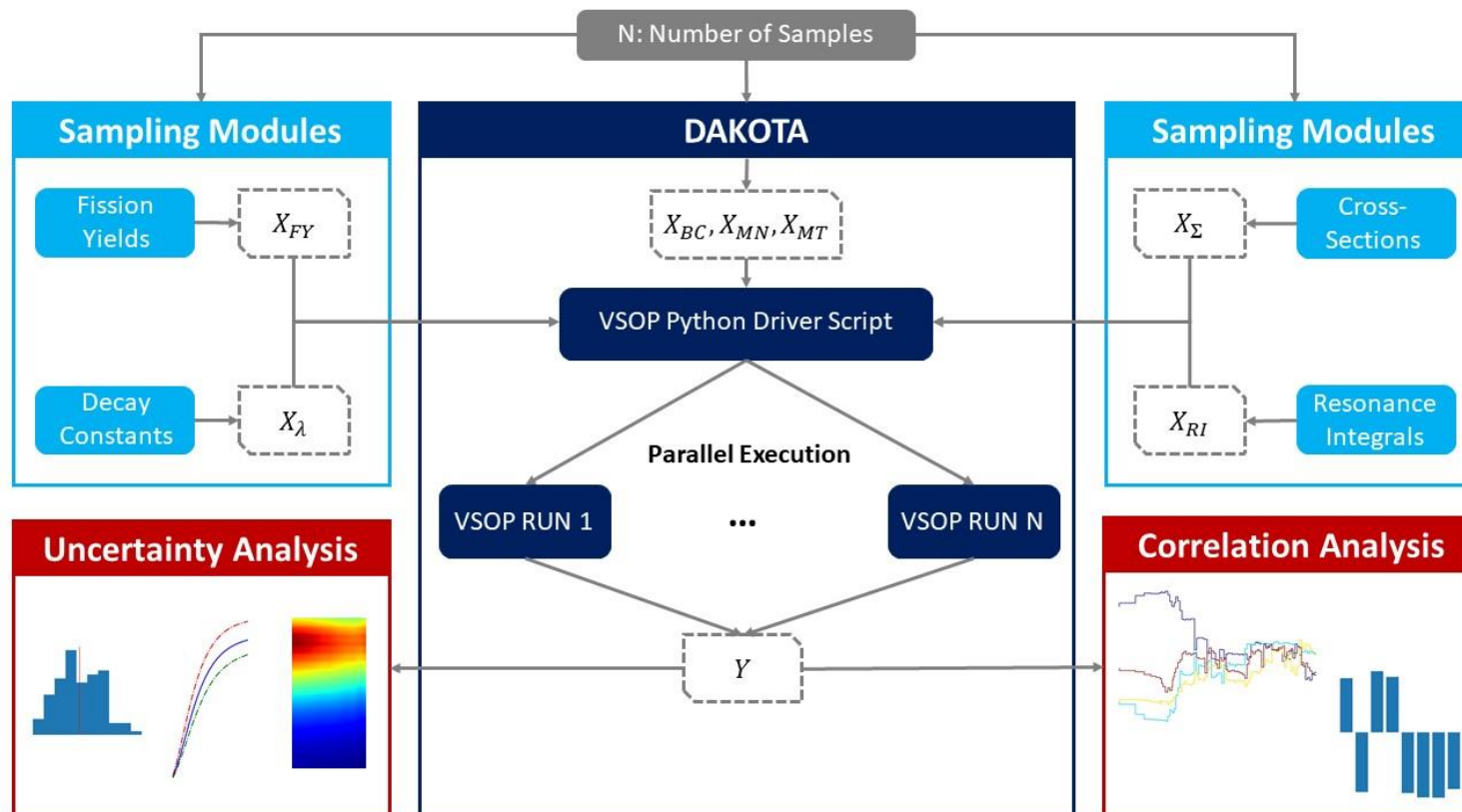
Input Sources of Uncertainties:

1. Manufacturing parameters (X_{MN}).
2. Boundary conditions (X_{BC}).
3. Material properties (X_{MT}).
4. Microscopic cross-sections (X_{Σ}).
5. Resonance integral parameters (X_{RI}).
6. Fission yields (X_{FY}).
7. Decay constants (X_{λ}).



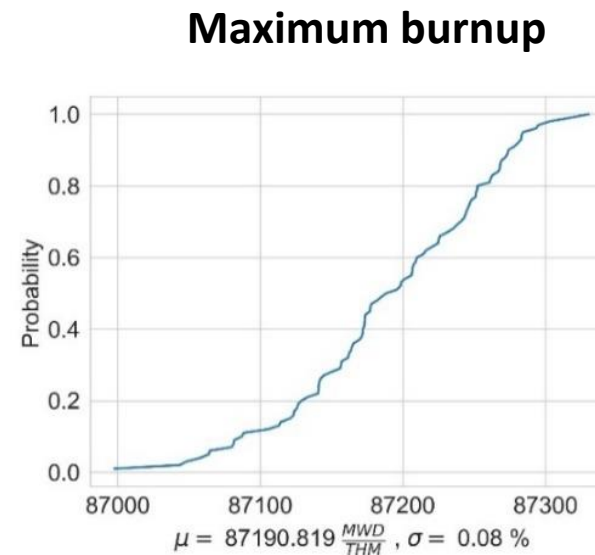
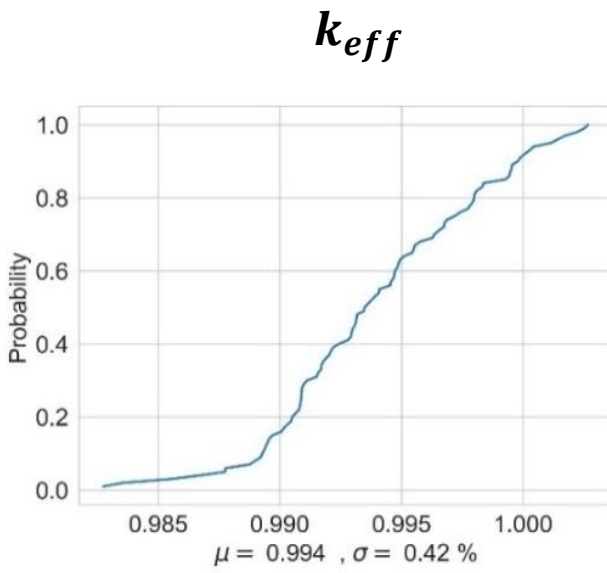
Output quantities of interest and analyses:

- Scalars (e.g. k_{eff})
- 1D/2D/3D functionals (e.g. gas temperature).
- Estimate statistical moments: mean, standard deviation.
- Histograms, Scatterplots and Empirical Cumulative Density Function (ECDF).
- Estimation of Pearson and Spearman correlations.

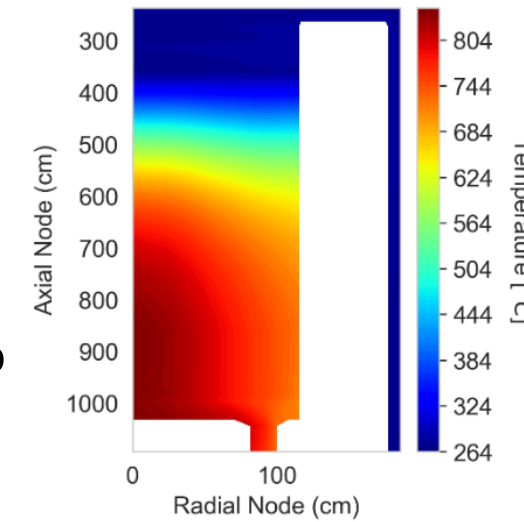


NCSU MULTI-PHYSICS PLATFORM FOR UNCERTAINTY QUANTIFICATION

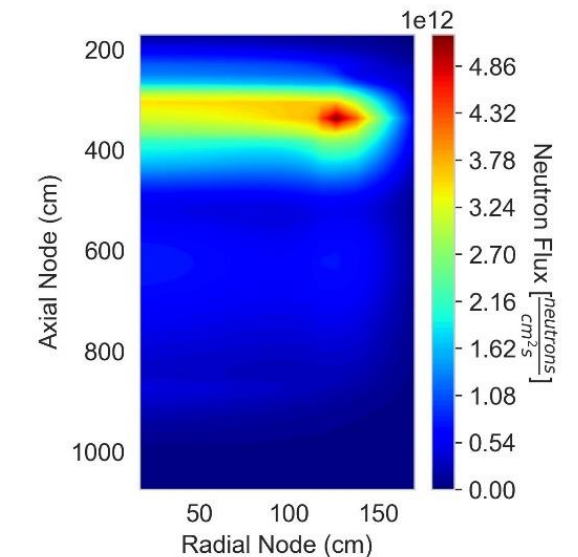
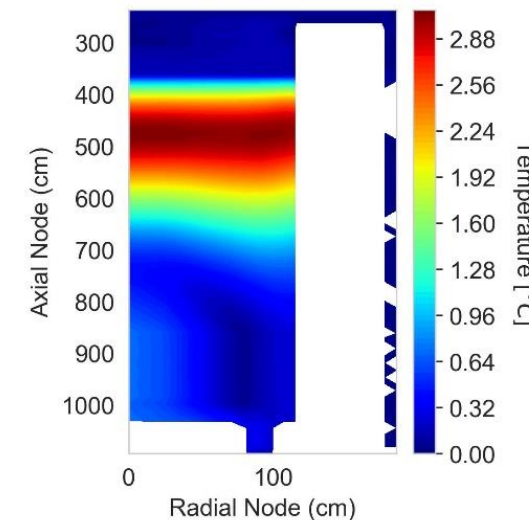
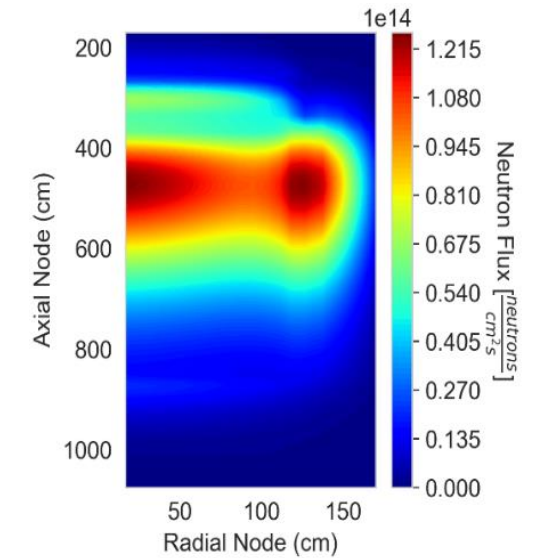
- 80 MWth PB-HTGR
- Around 450,000 fuel pebbles at 61% packing fraction.
- Around 15,000 Triso particles with UO_2 kernels of 10.5% enrichment.
- $k_{eff}=0.99467$, maximum burnup is 87204 MWD/THM.
- Uncertainty analysis results for the k_{eff} , maximum burnup
- The standard deviation of the k_{eff} is ~ 420 pcm, which is close to uncertainty ranges found in literature.



Gas Temperature



Thermal Flux



SUMMARY - THE DEMAND?

- Cost per computational unit continues to drop in dollars and time.
- End customer continues to demand better simulations.
- Direct 3D heterogeneous neutron transport solutions coupled with CFD are still cost prohibitive and data intensive for industry (routine) calculations.
- Iterative (embedded) neutronics methodologies, such as ITDM, coupled with improved sub-channel codes are becoming attractive alternatives.
- The multi-physics coupling is being successfully extended to include detail fuel modeling in reactor cores.
- Uncertainty and sensitivity analysis methods need to be considered as an integral part in the development of multi-physics methods.
- Comprehensive benchmark framework is needed to verify/validate sensitivity and uncertainty analysis methods for multi-physics applications.

SUMMARY - THE GOAL?

Advanced M&S Tools which:

- Leverage existing state-of-the-art single-physics tools to solve coupled multi-physics nuclear reactor problems.
- Solve high-fidelity reactor transient problems in a manner closer to first principles:
 - ✓ *Difficult-to-model multi-physics phenomena;*
 - ✓ *Large problem sizes that can't be handled with conventional codes.*
- Provide insight into physical phenomena that can't be easily measured or accounted for in conventional tools.

Next Class: *Verification, Validation, and Uncertainty Quantification of Reactor Multi-Physics Modeling*